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Huang et al.

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(54) **CHIP STRUCTURE WITH CONDUCTIVE PILLAR AND METHOD FOR FORMING THE SAME**

(58) **Field of Classification Search**
CPC H01L 24/13; H01L 23/5226; H01L 24/11
See application file for complete search history.

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Primary Examiner — Laura M Menz

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(74) *Attorney, Agent, or Firm* — McClure, Qualey & Rodack, LLP

(65) **Prior Publication Data**
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(57) **ABSTRACT**

A chip structure is provided. The chip structure includes a substrate. The chip structure includes a first conductive line over the substrate. The chip structure includes an insulating layer over the substrate and the first conductive line. The chip structure includes a conductive pillar over and passing through the insulating layer. The conductive pillar is formed in one piece, the conductive pillar is in direct contact with the first conductive line, and a first sidewall of the first conductive line extends across a second sidewall of the conductive pillar in a top view of the first conductive line and the conductive pillar. The chip structure includes a solder bump on the conductive pillar. The solder bump is in direct contact with the conductive pillar.

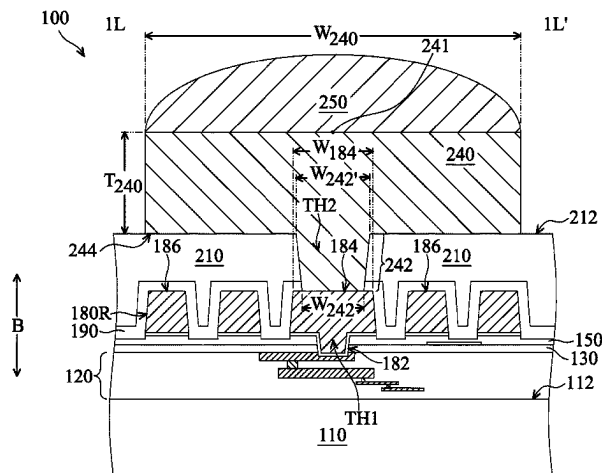
Related U.S. Application Data

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H01L 23/00 (2006.01)
H01L 23/522 (2006.01)

(52) **U.S. Cl.**
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20 Claims, 30 Drawing Sheets



(52) U.S. Cl.

CPC H01L 2224/1147 (2013.01); H01L
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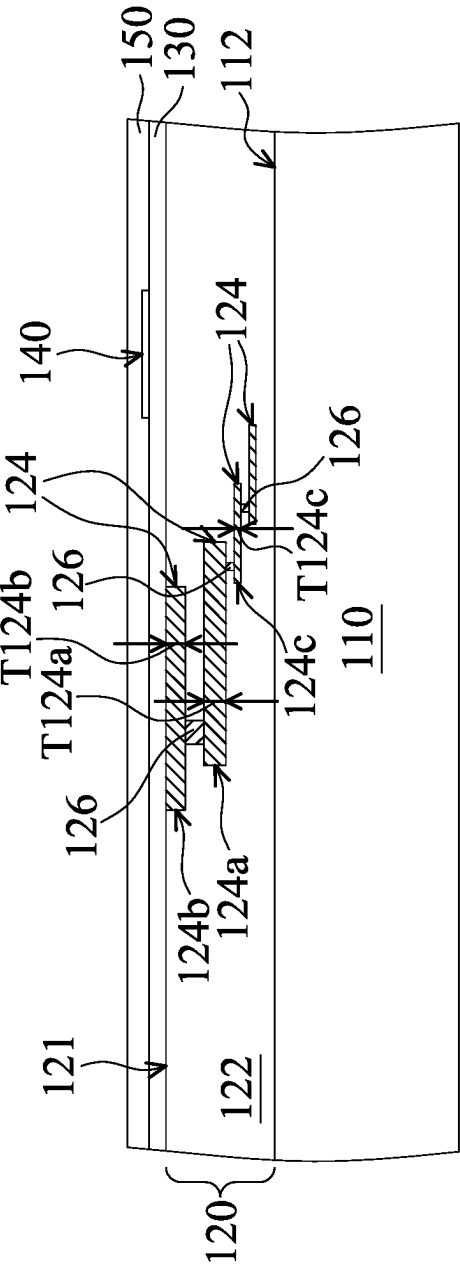


FIG. 1A

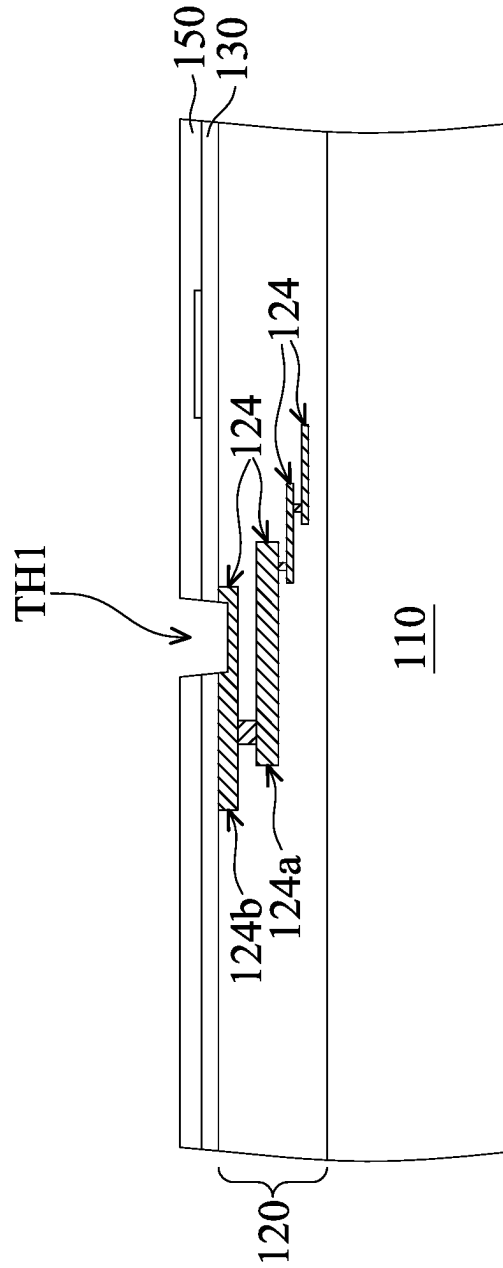


FIG. 1B

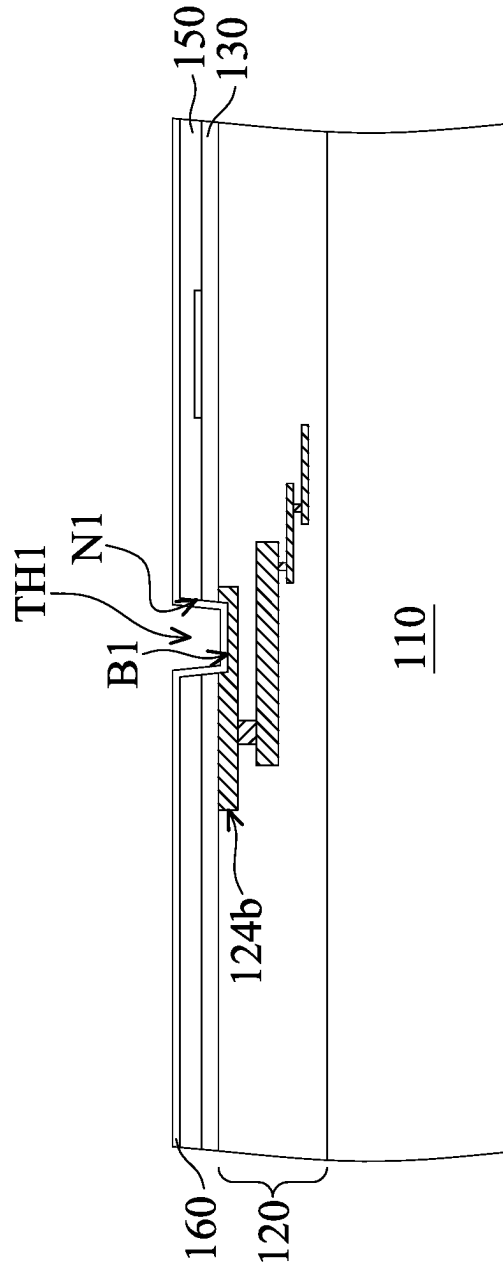


FIG. 1C

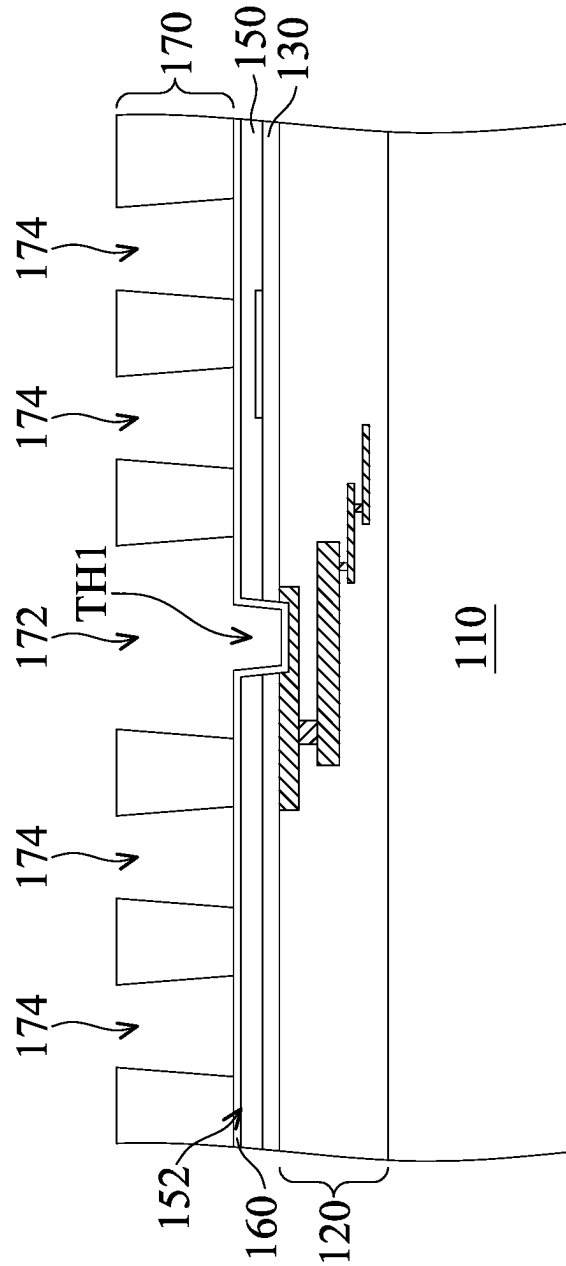


FIG. 1D

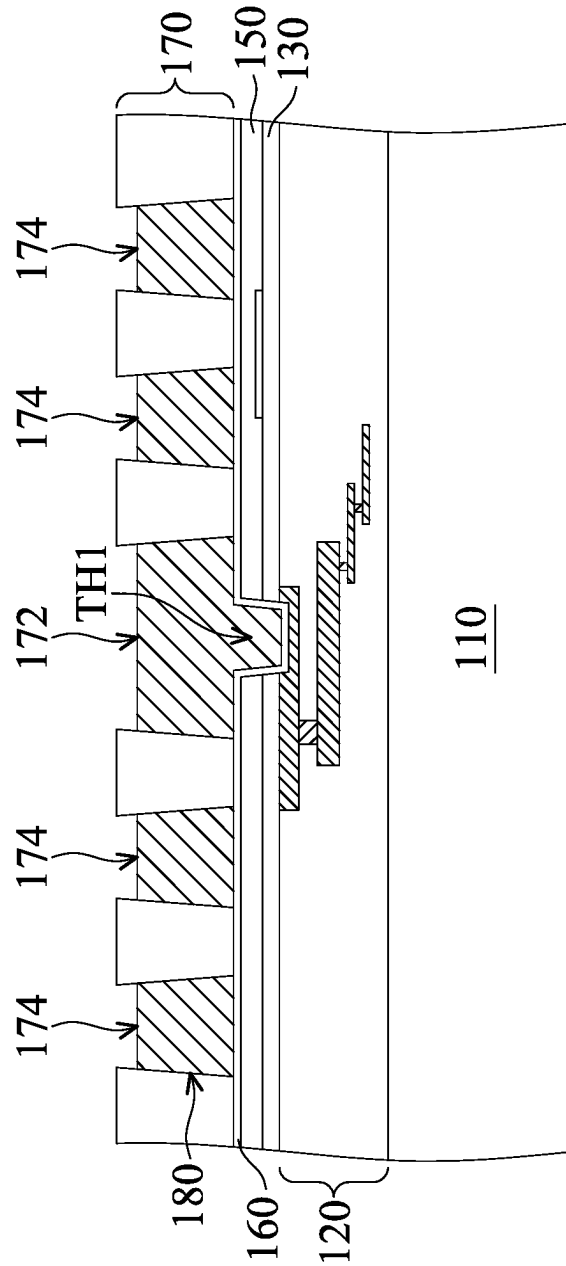


FIG. 1E

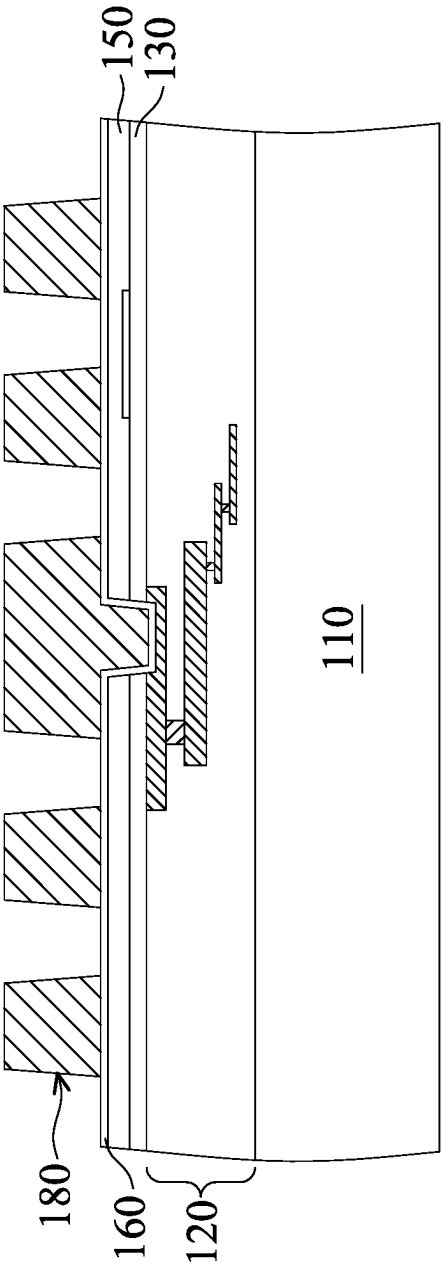


FIG. 1F

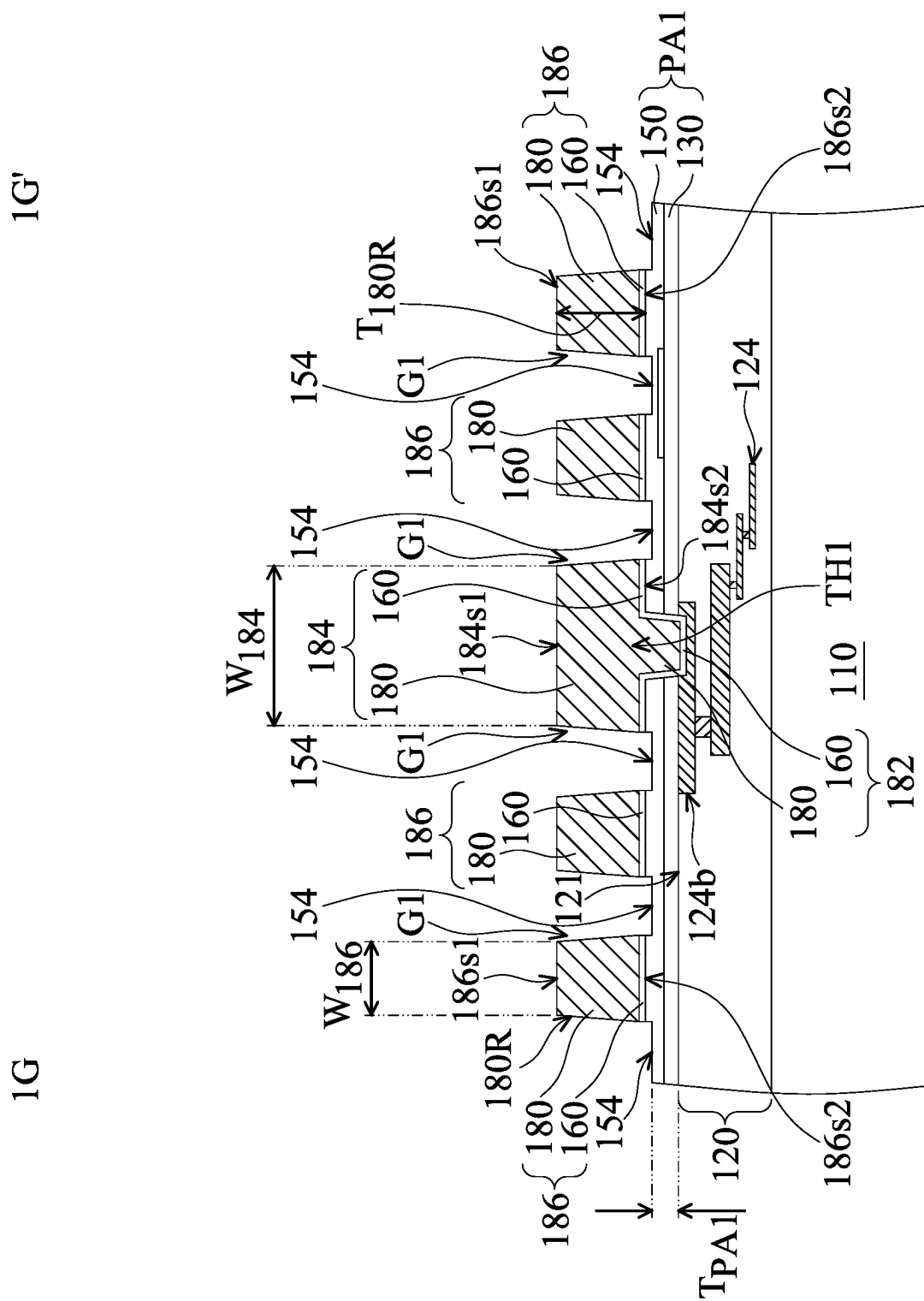


FIG. 1G

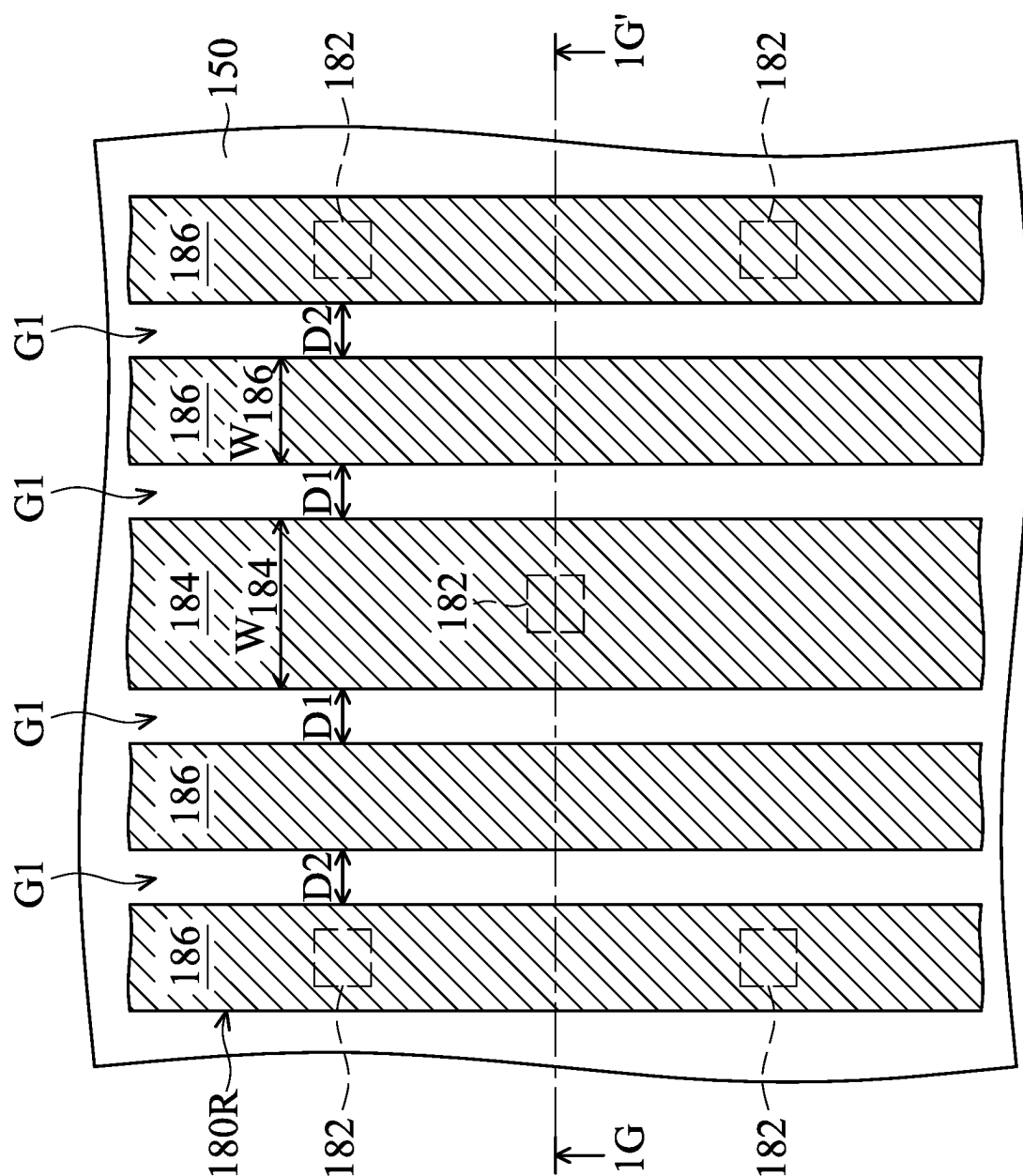


FIG. 1G-1

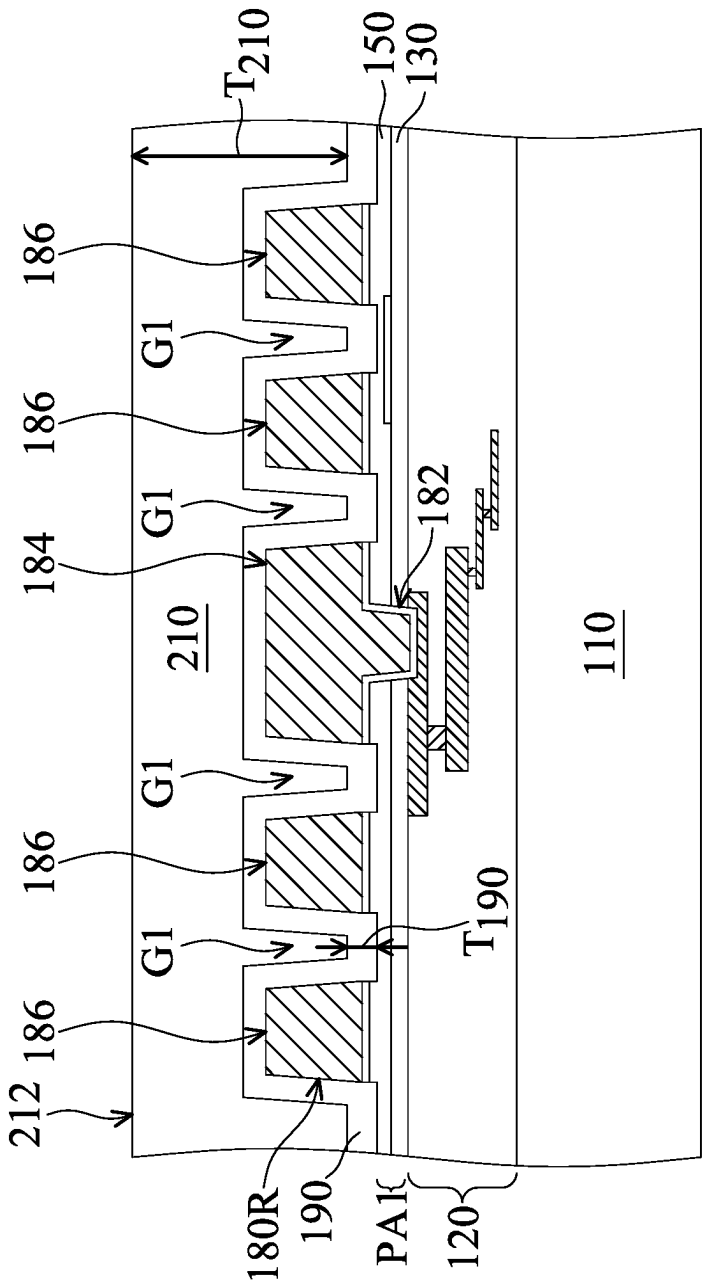


FIG. 1H

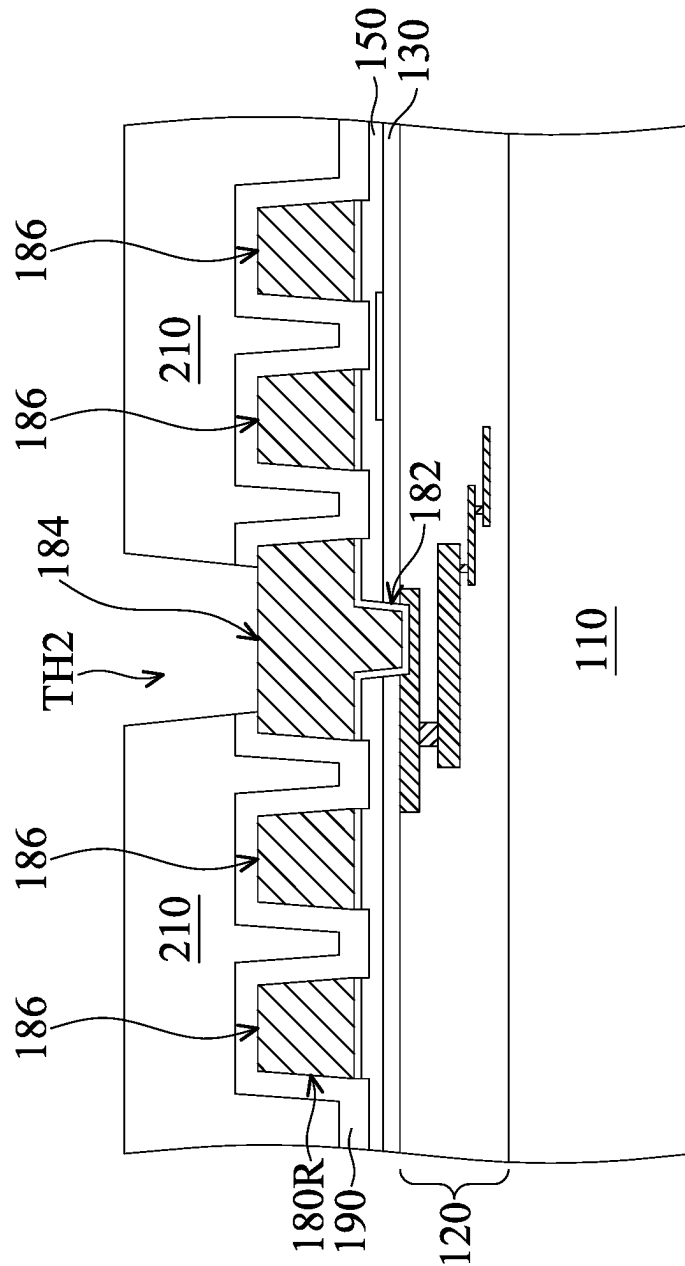


FIG. 11

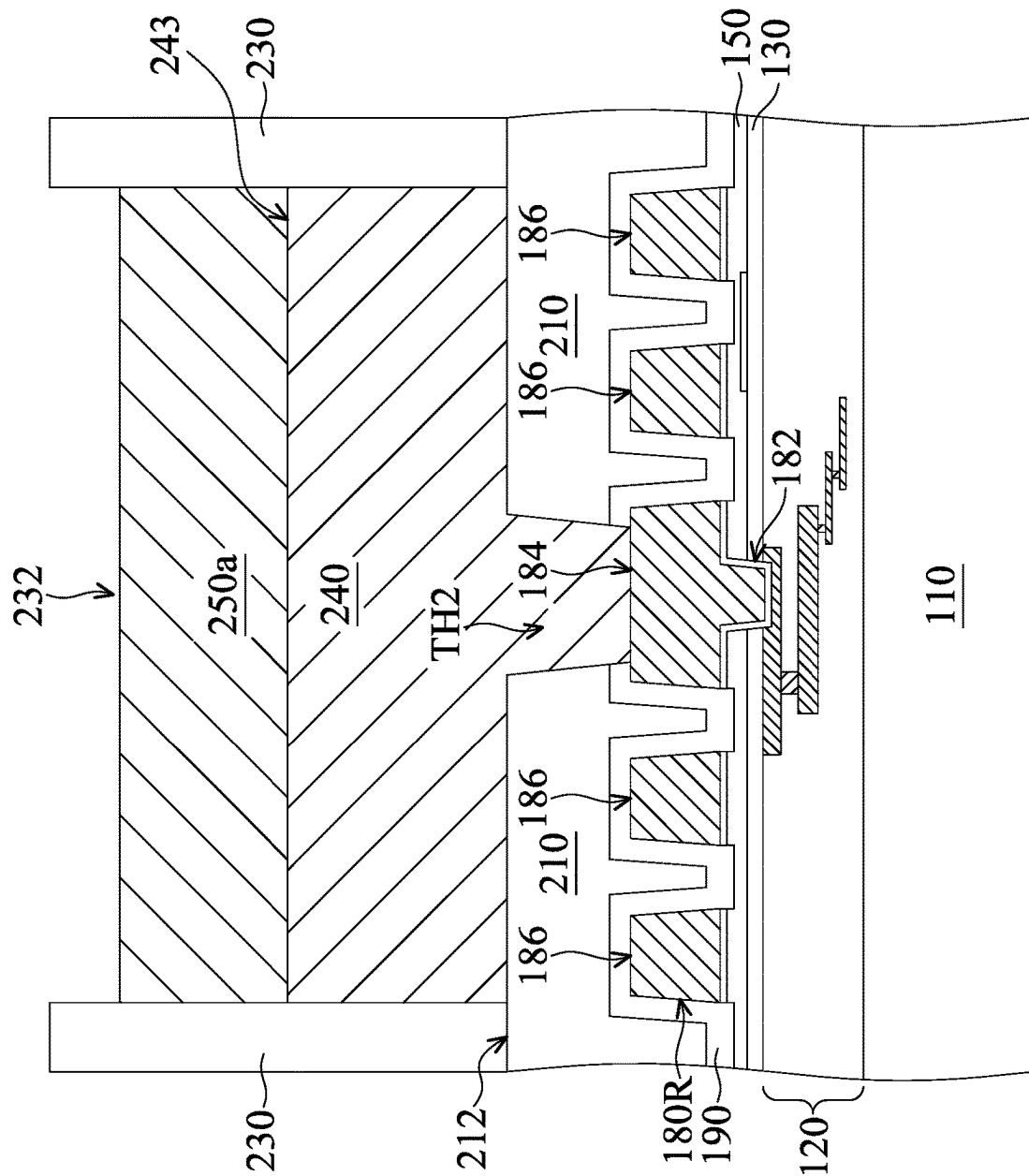


FIG. 1J

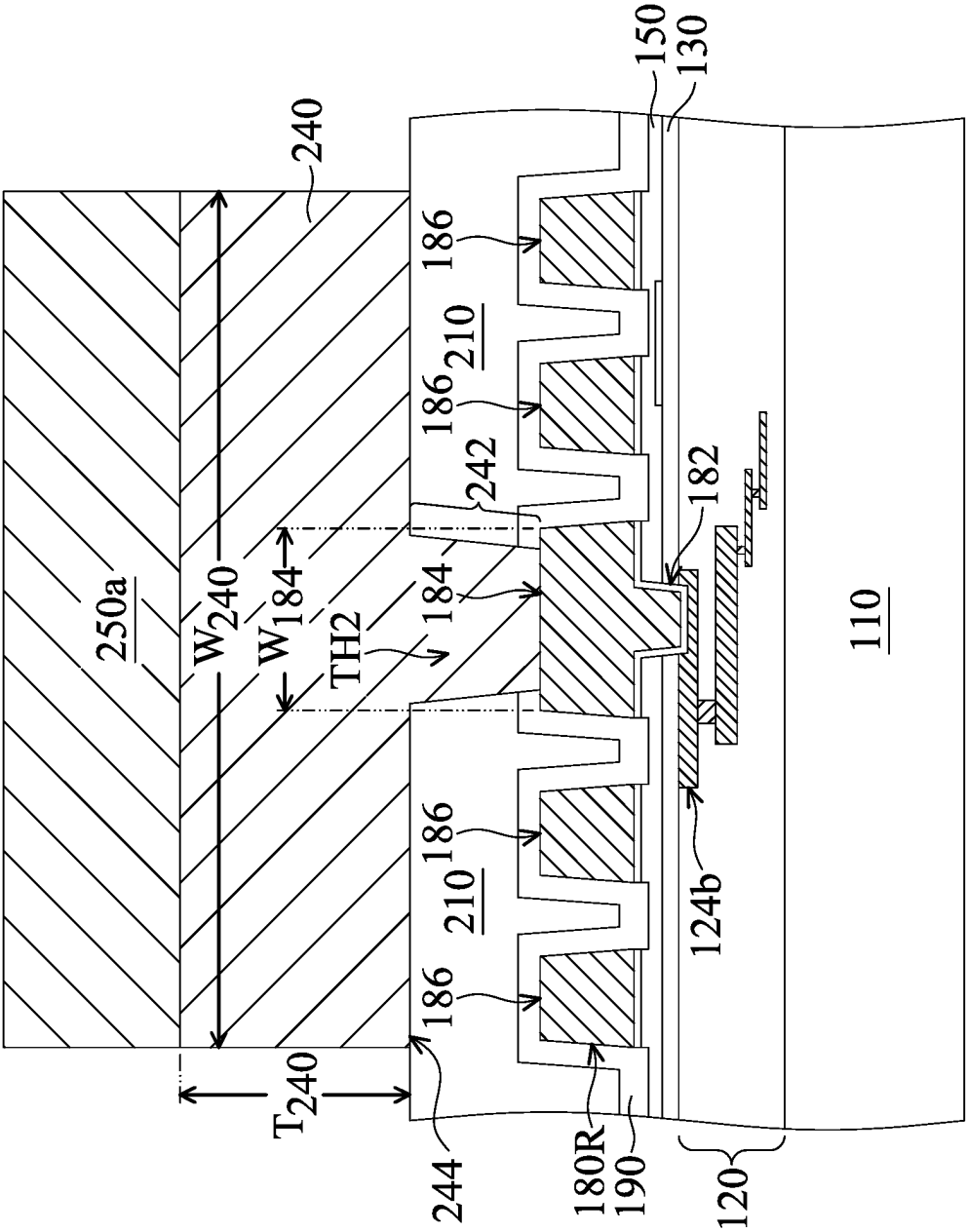


FIG. 1K

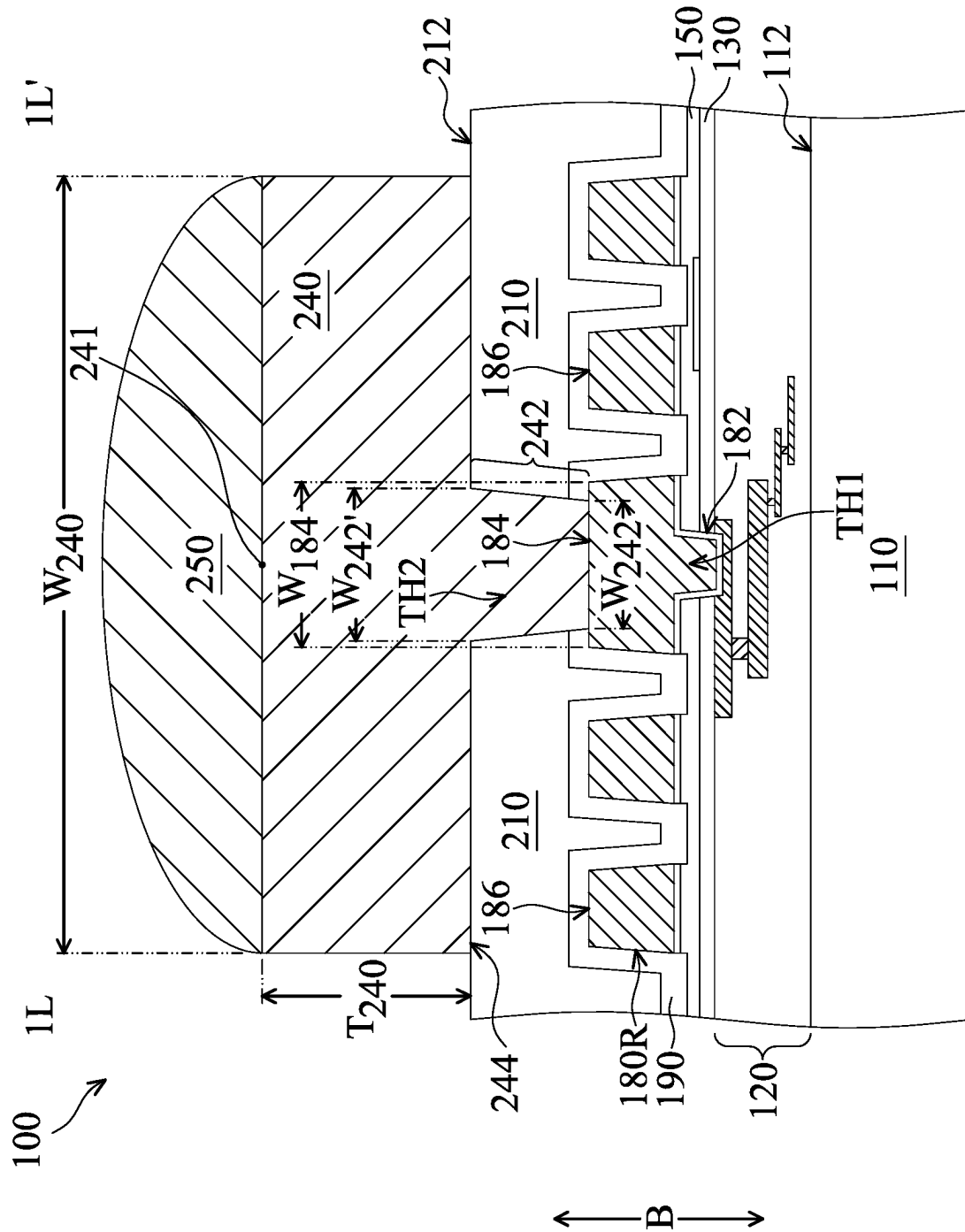


FIG. 1L

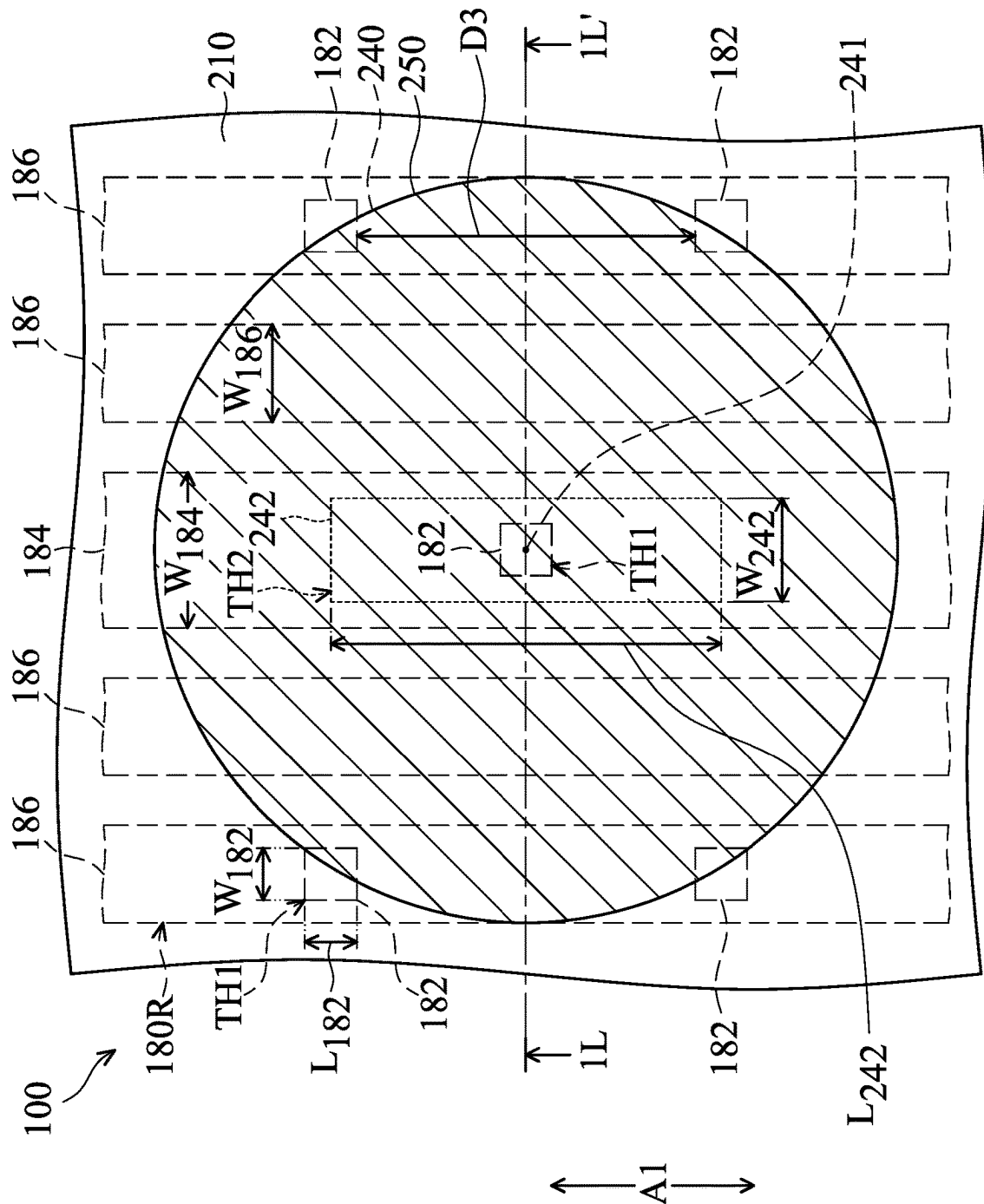


FIG. 1L-1

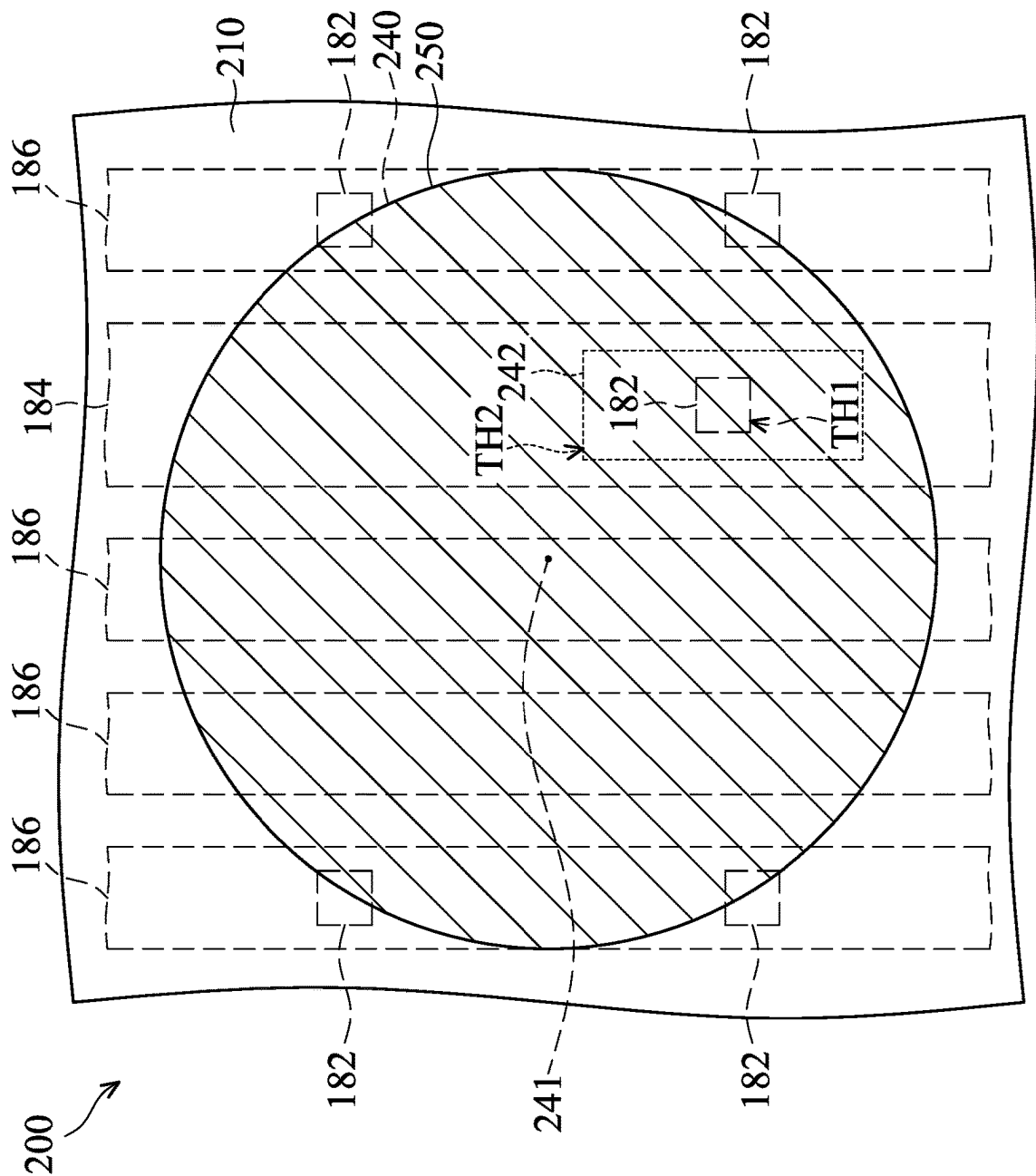


FIG. 2

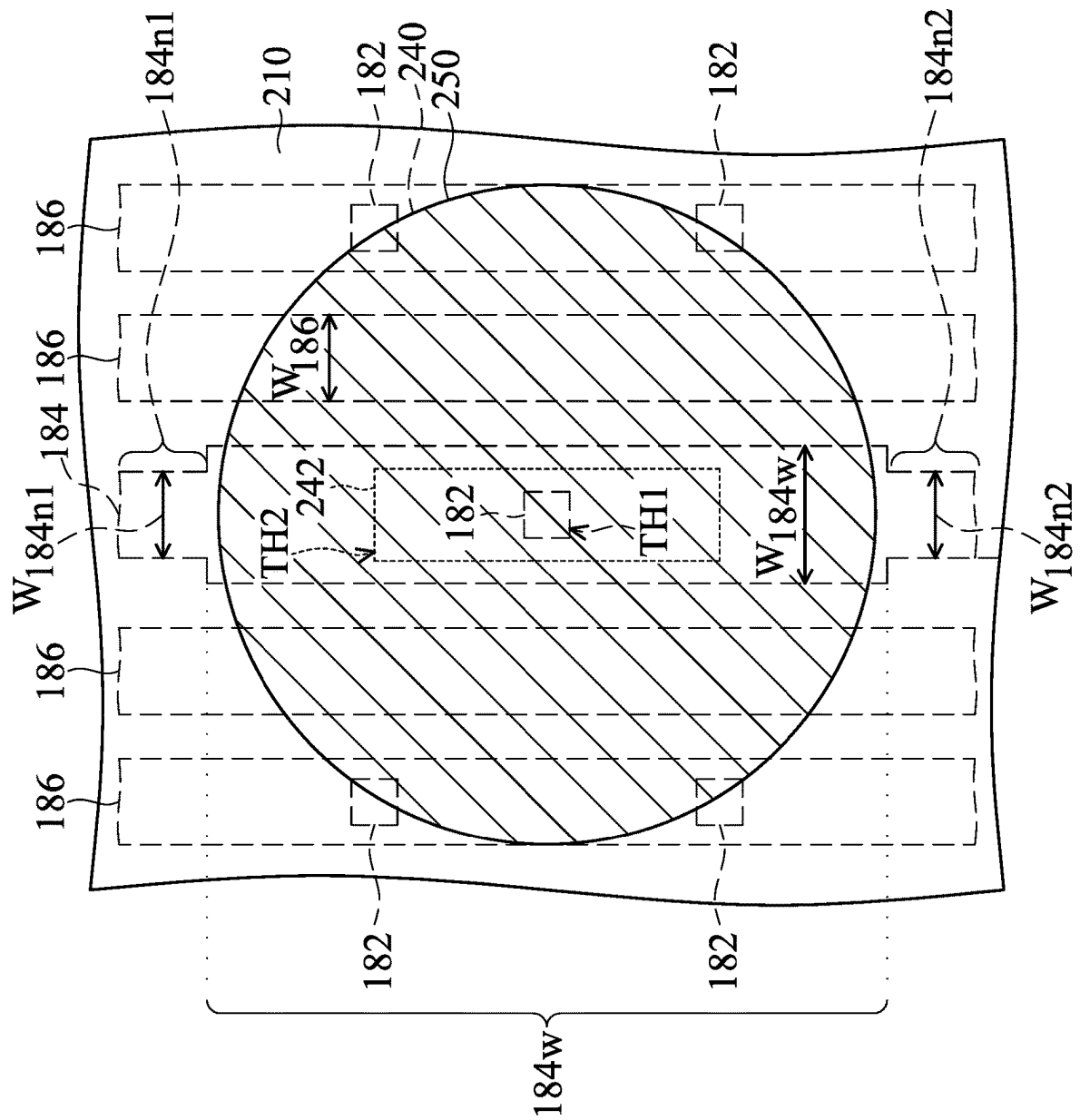


FIG. 3

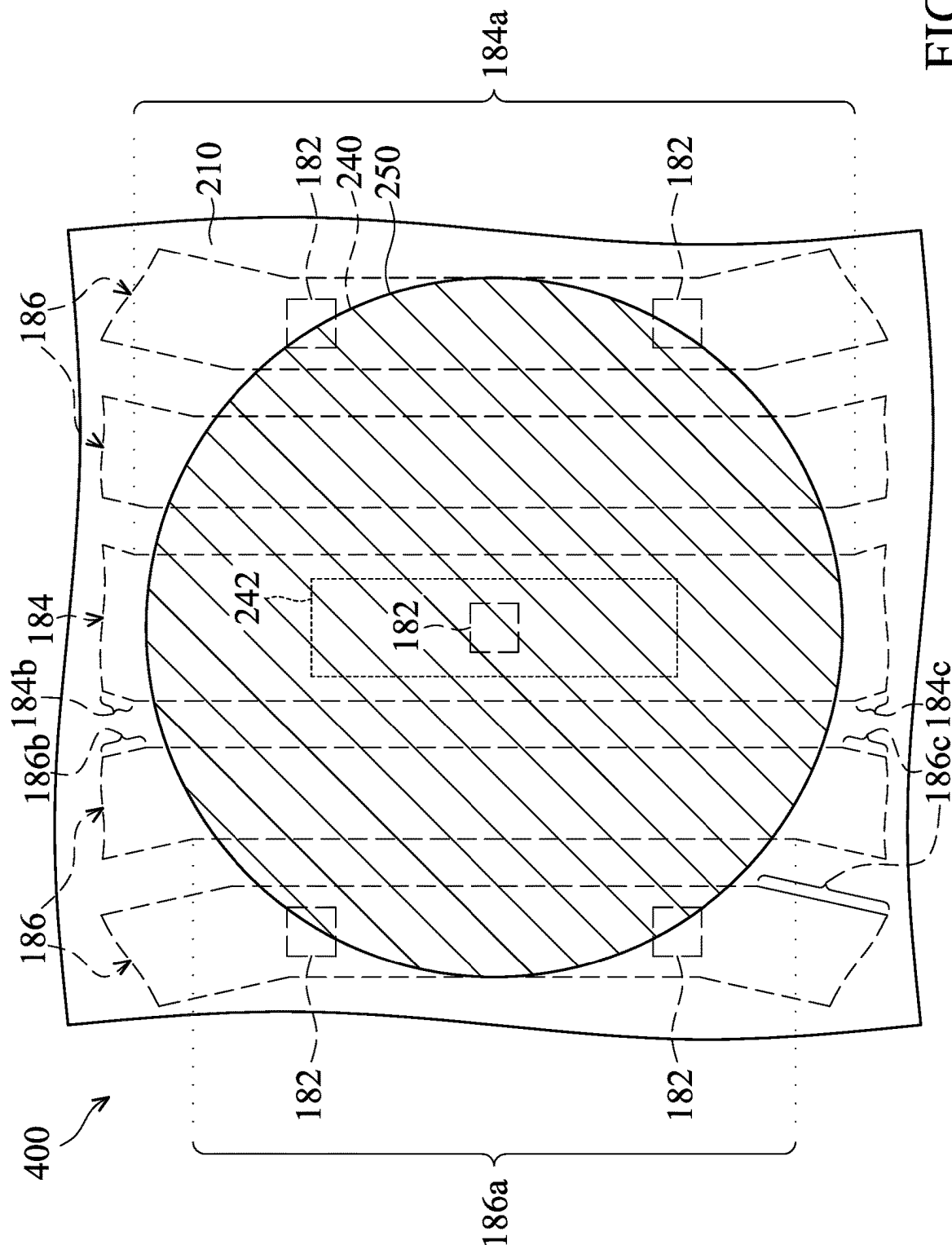


FIG. 4

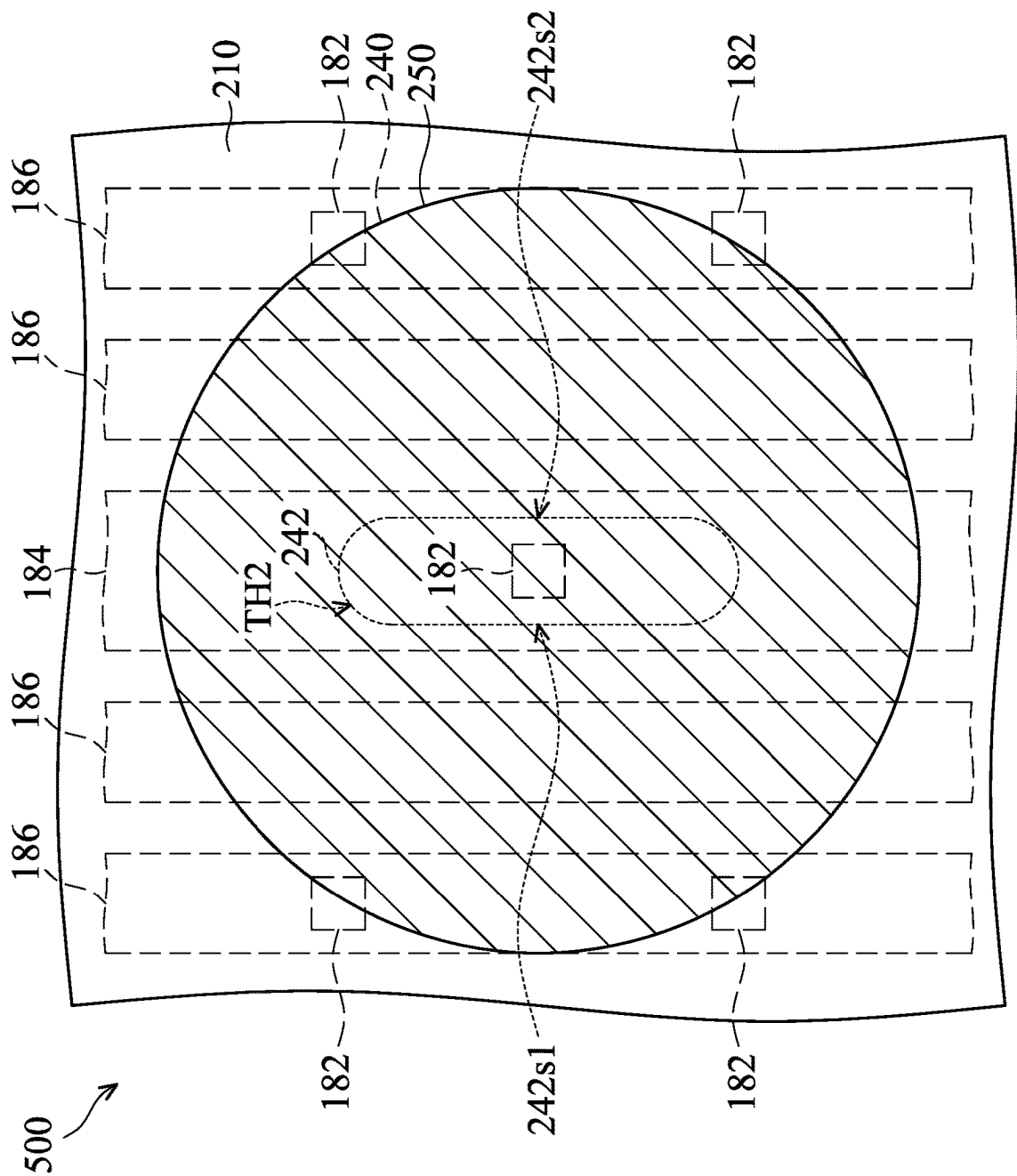


FIG. 5

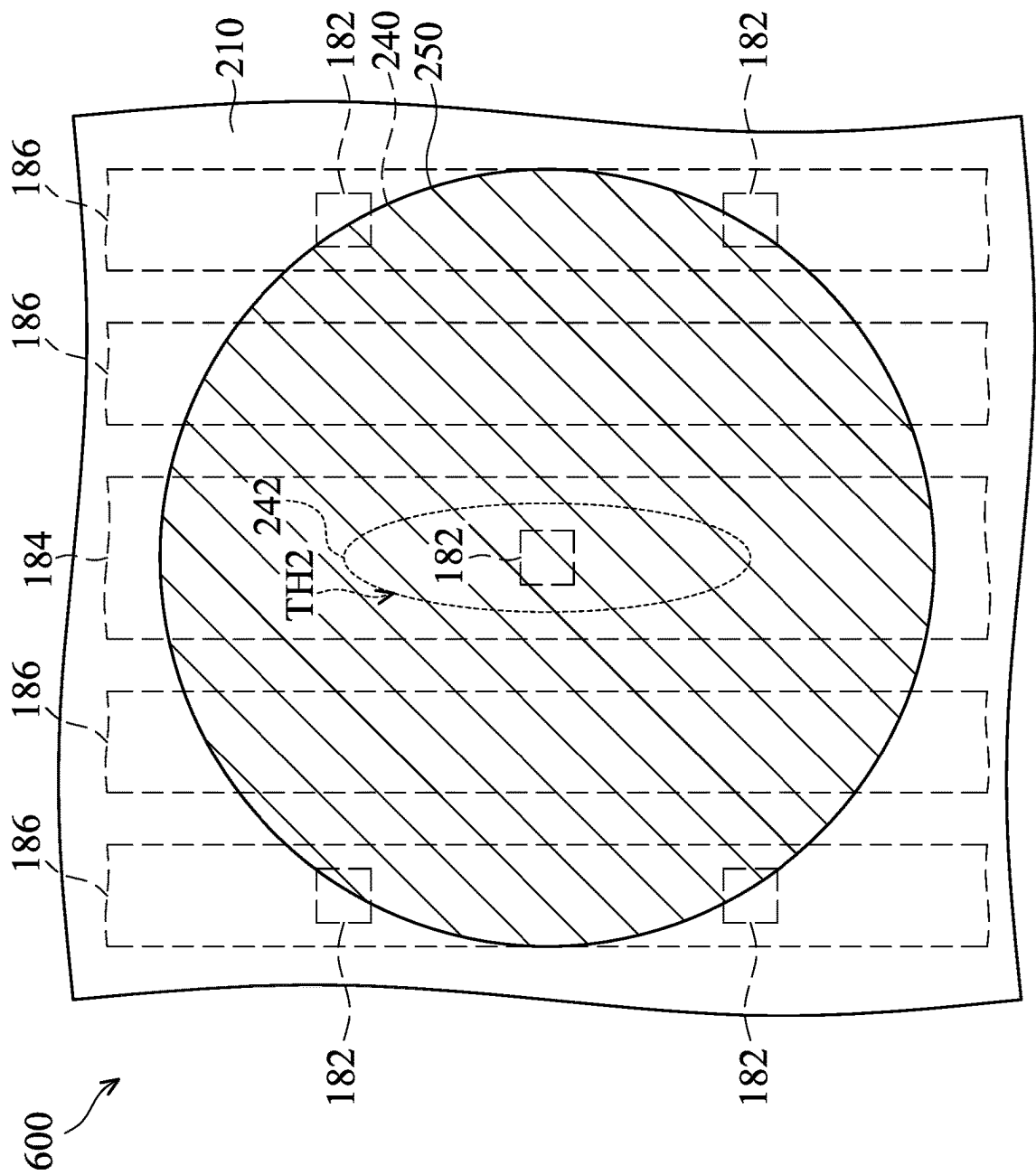


FIG. 6

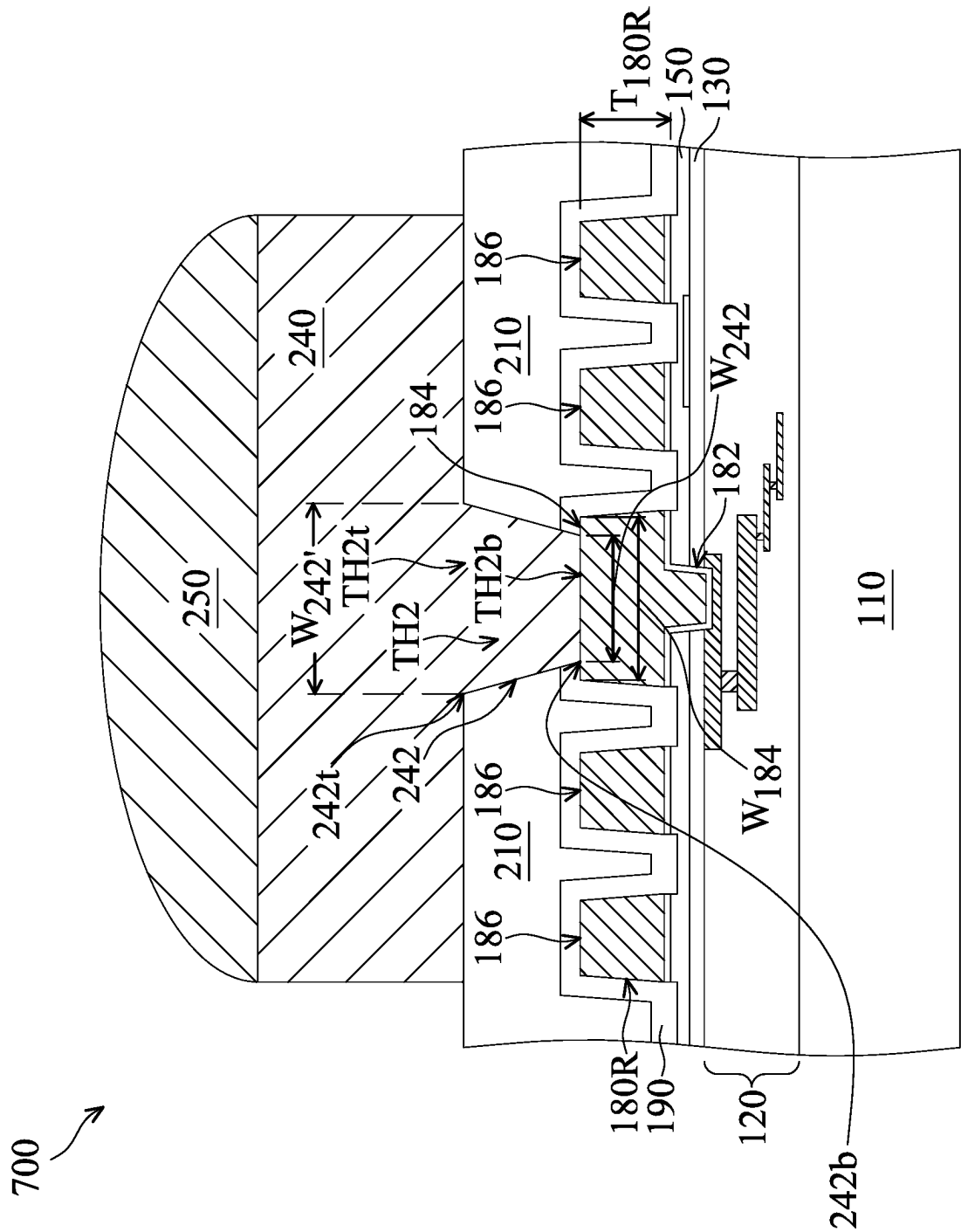


FIG. 7A

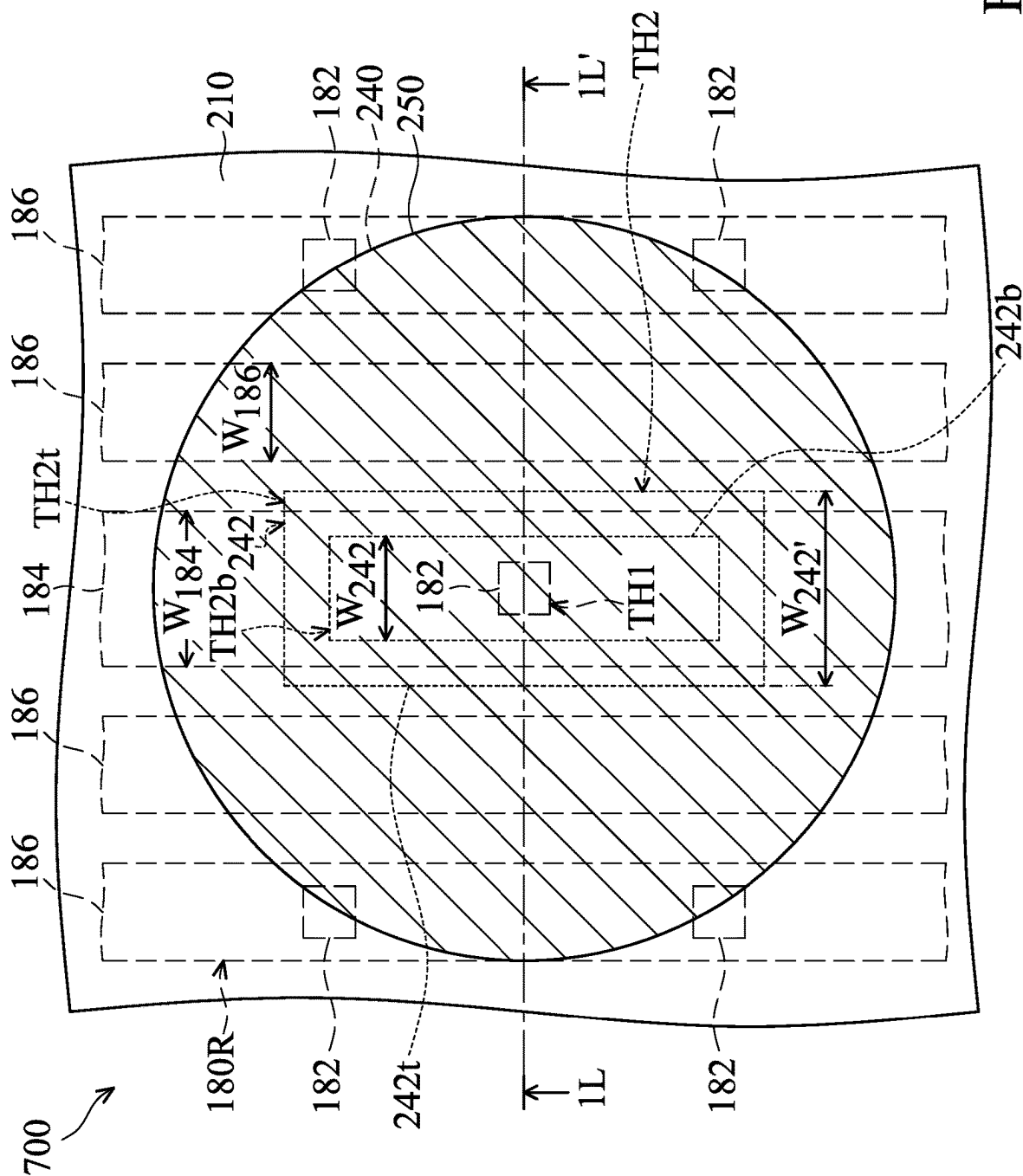


FIG. 7B

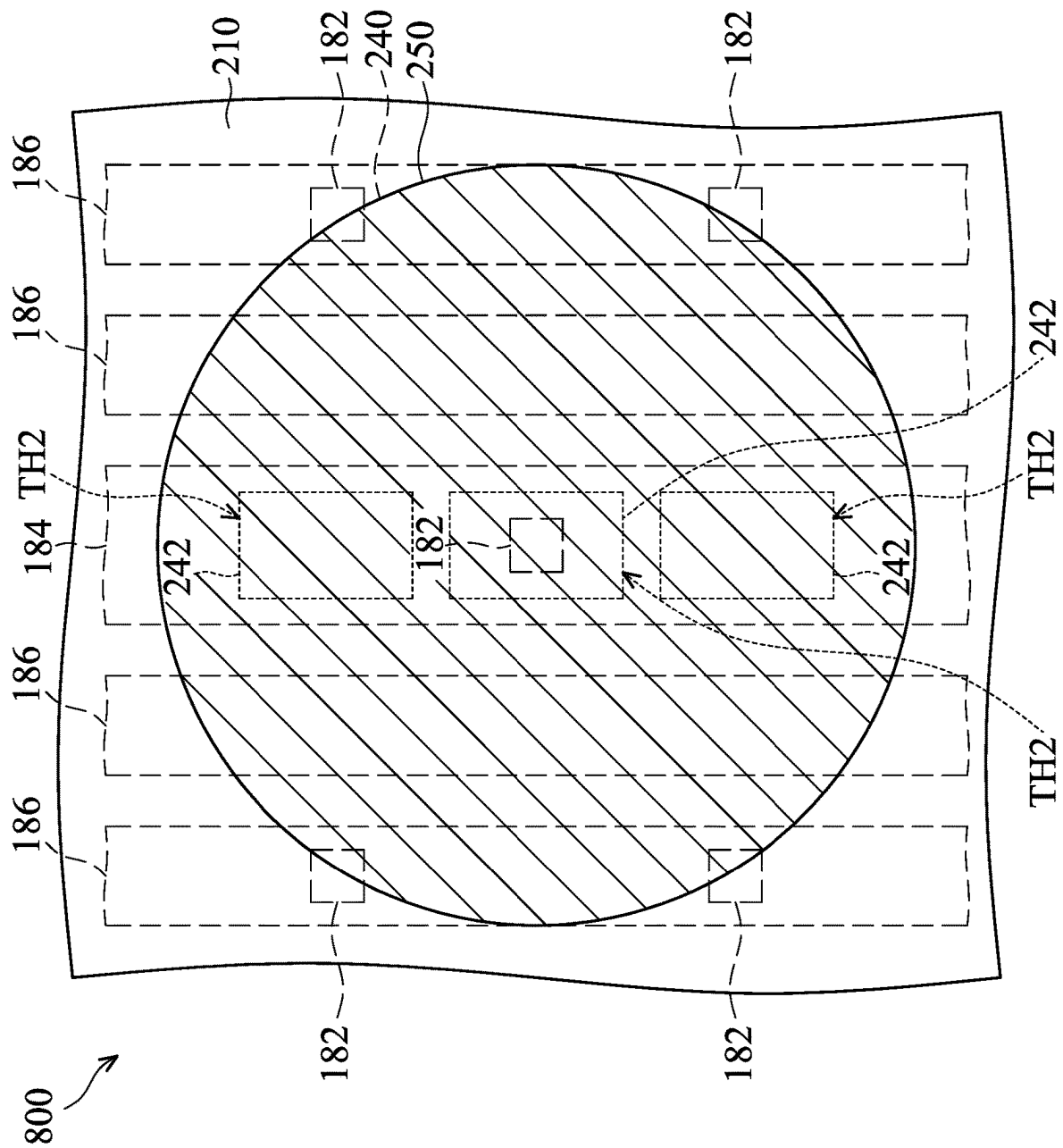
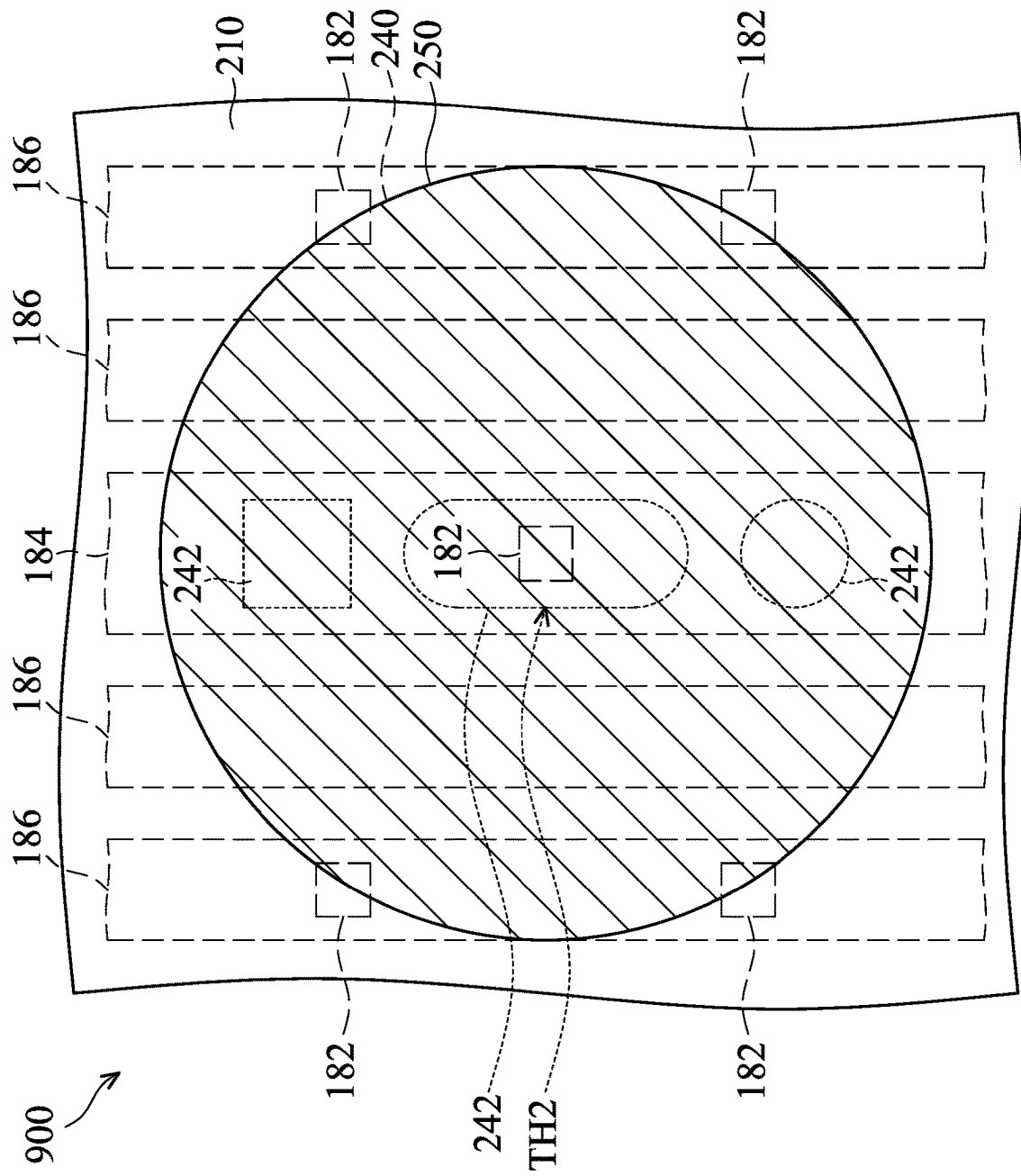


FIG. 8

FIG. 9



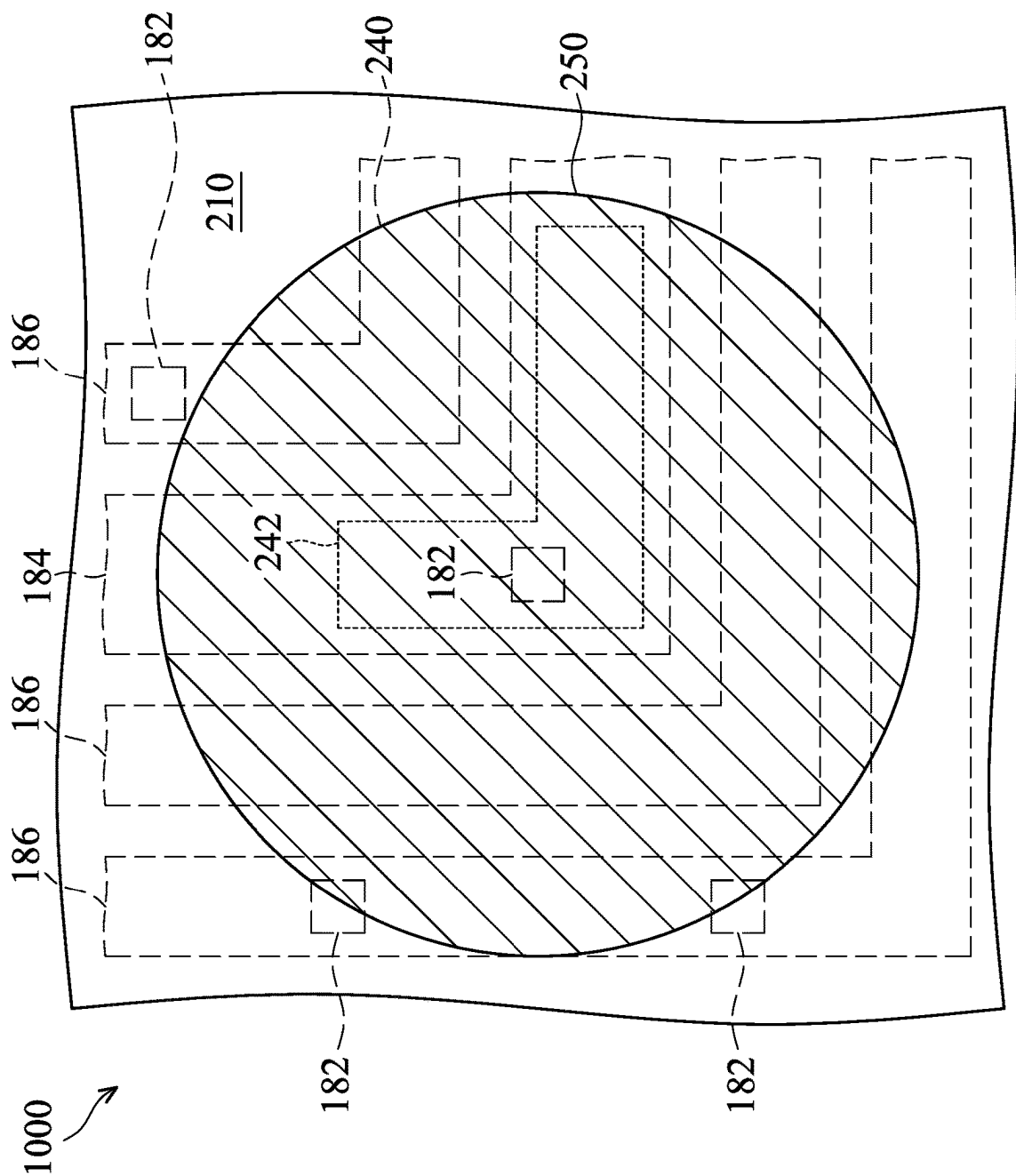


FIG. 10

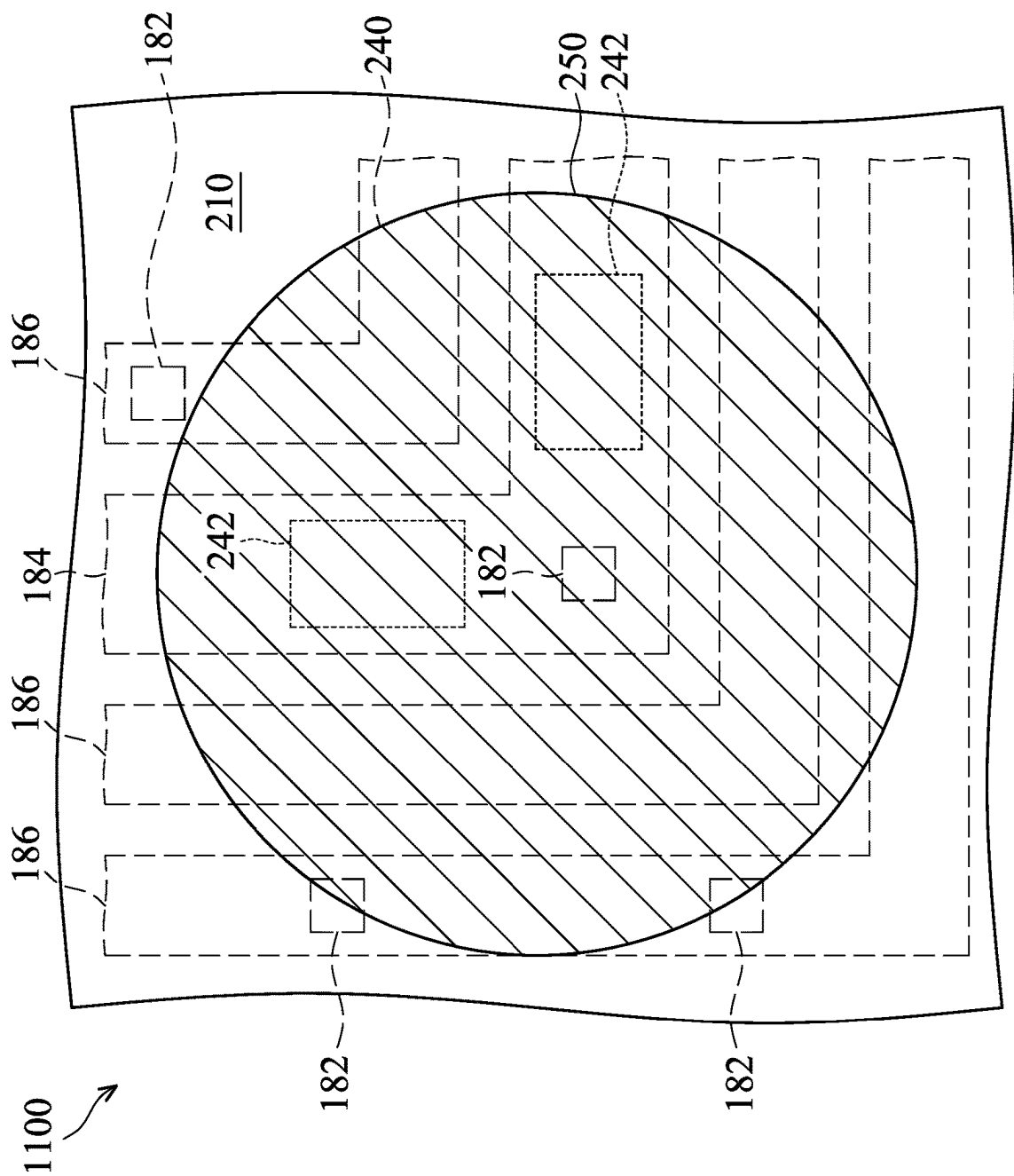


FIG. 11

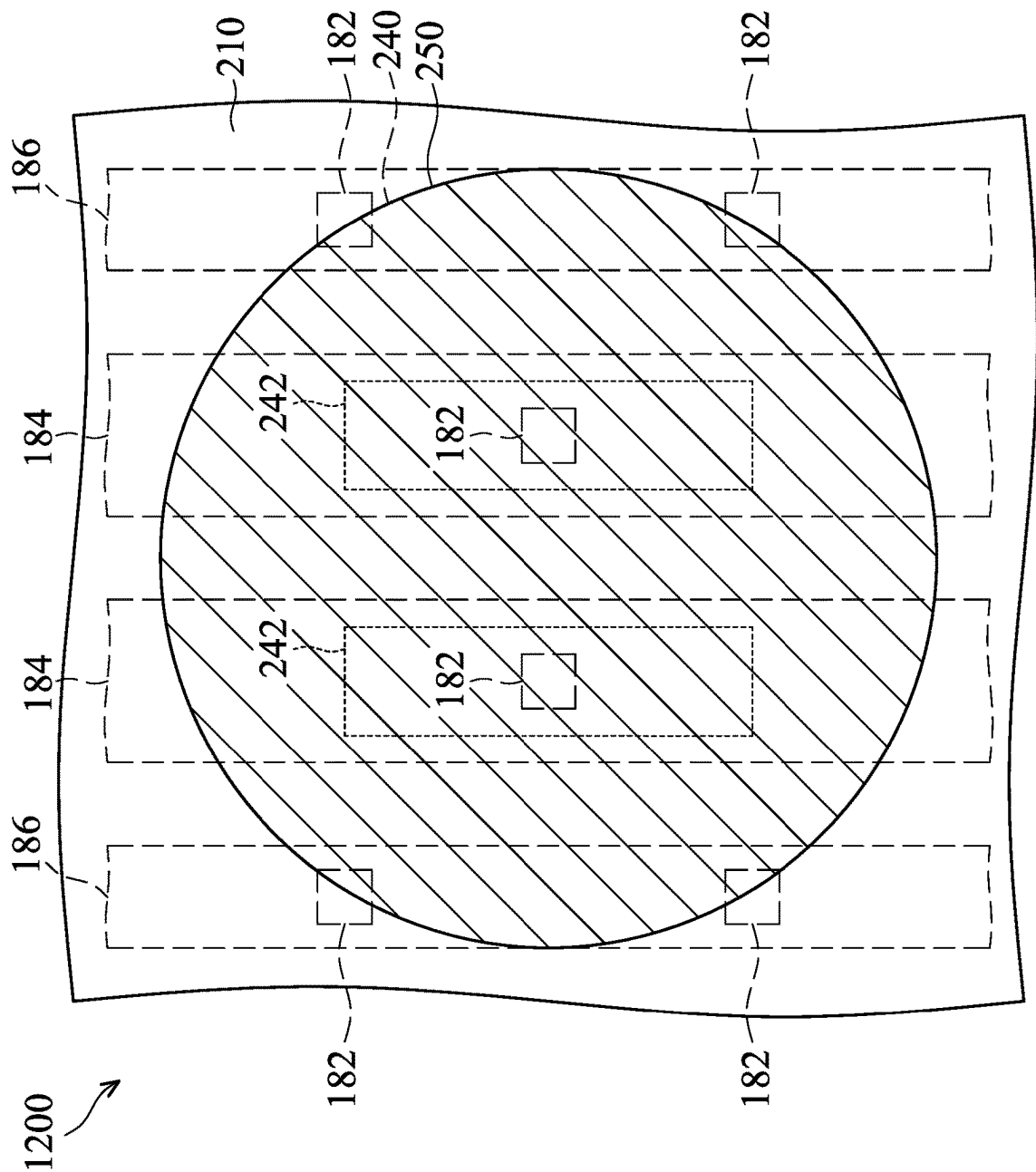


FIG. 12

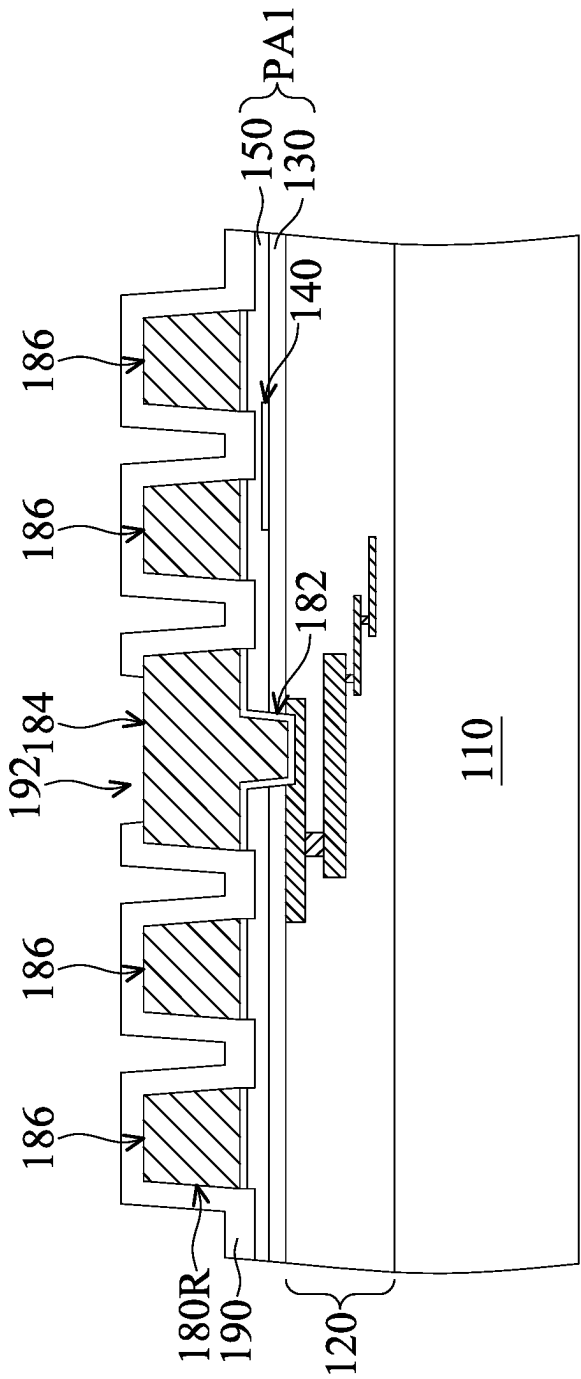


FIG. 13A

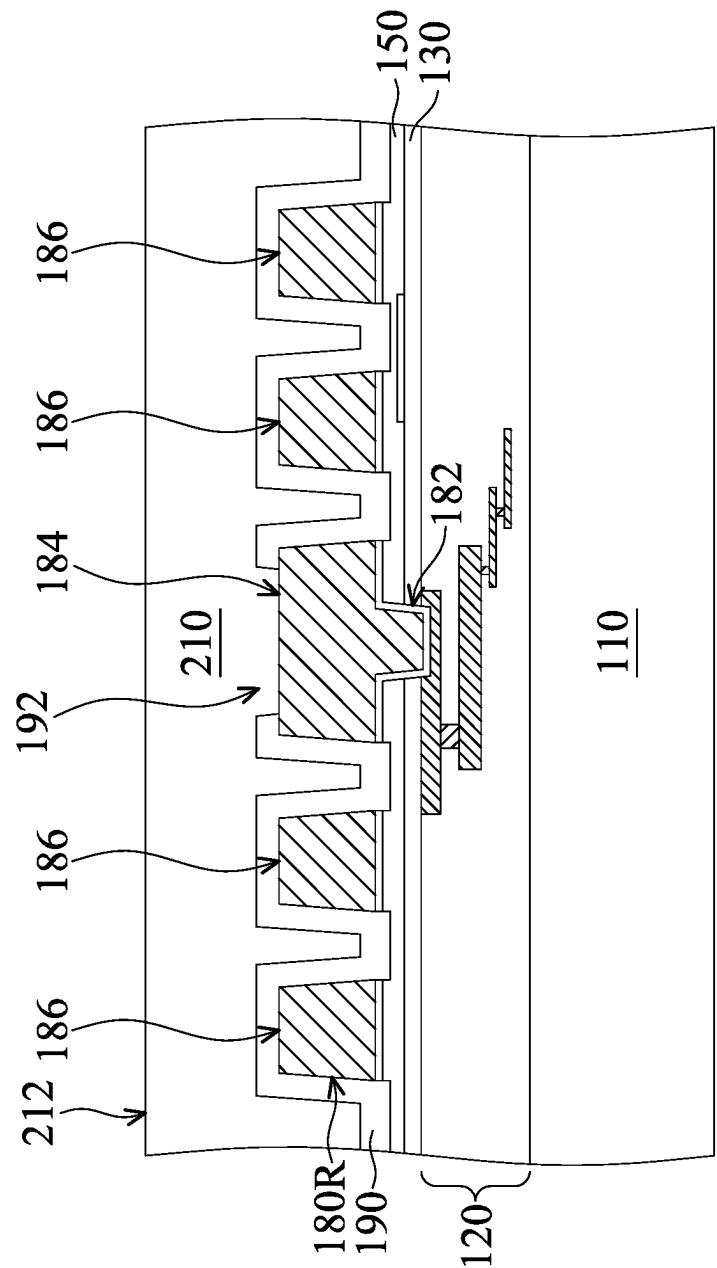


FIG. 13B

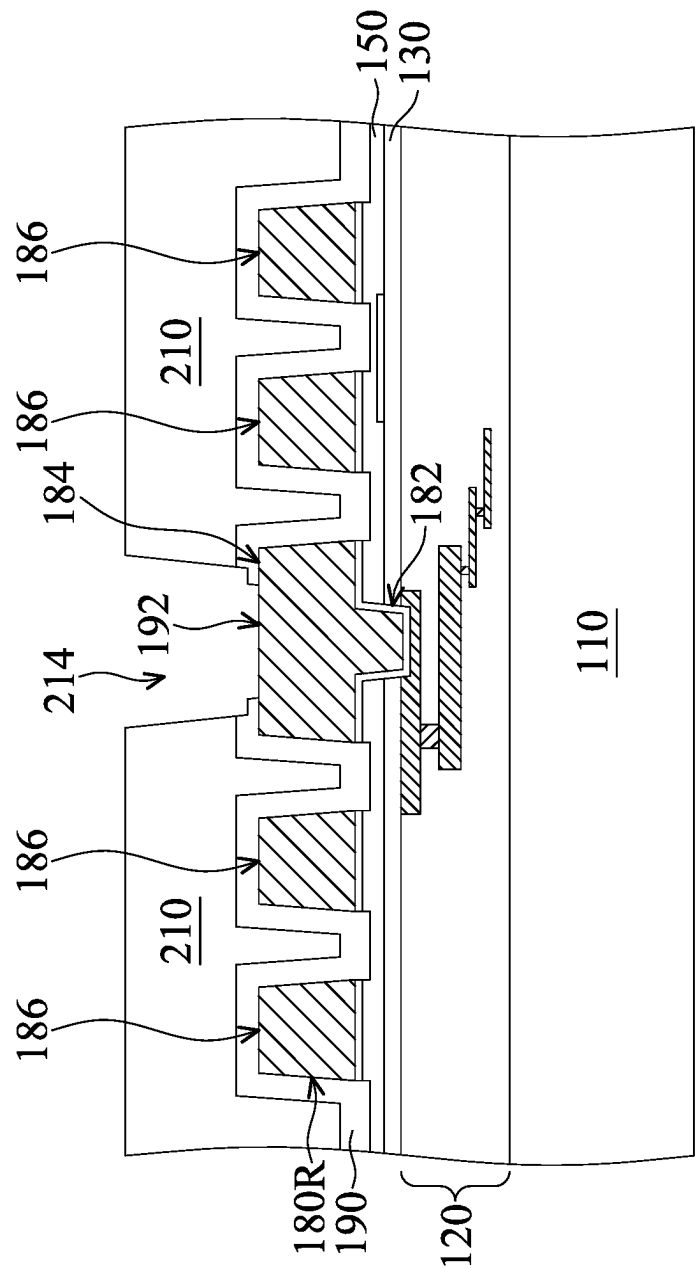


FIG. 13C

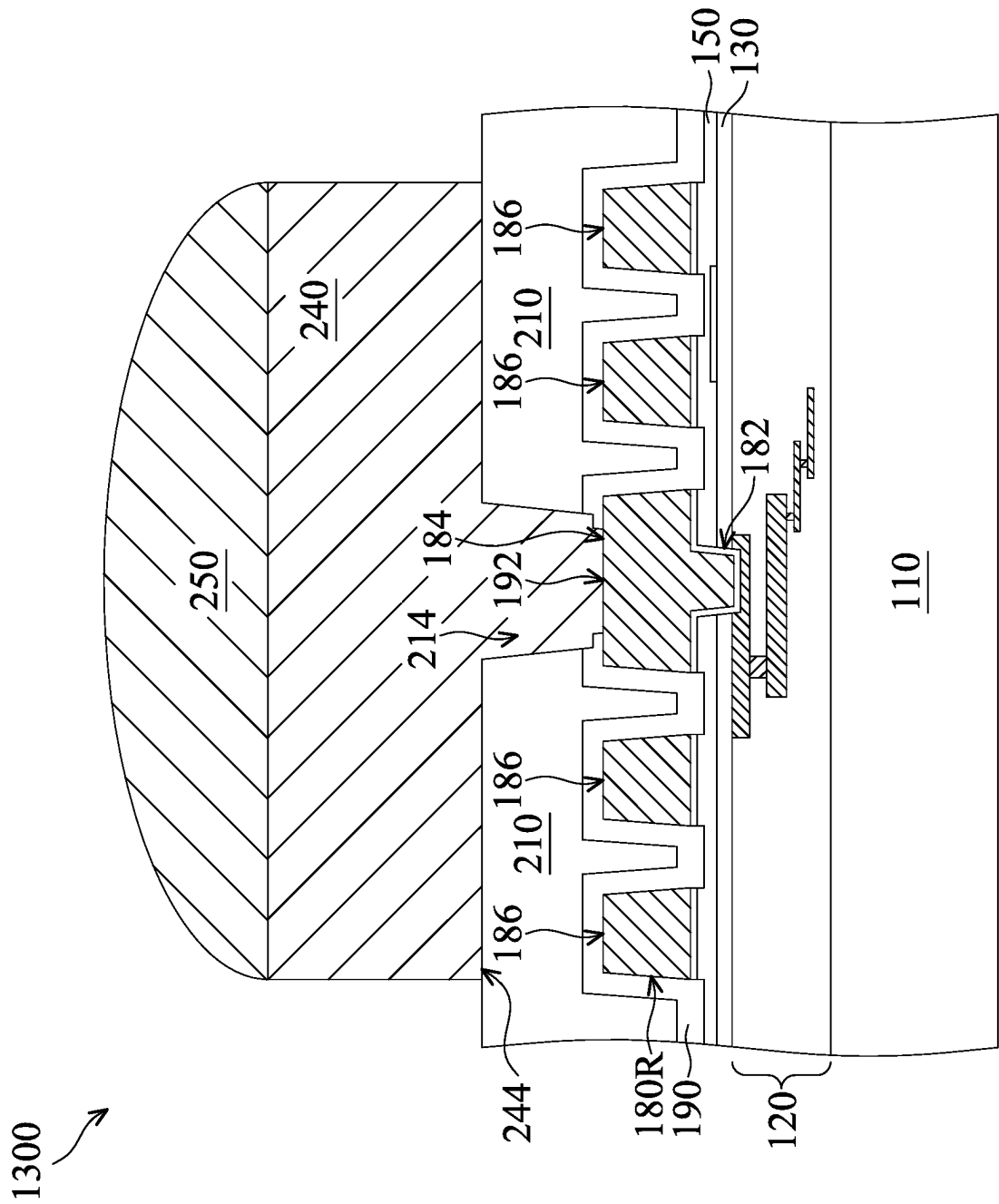


FIG. 13D

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CHIP STRUCTURE WITH CONDUCTIVE PILLAR AND METHOD FOR FORMING THE SAME

CROSS REFERENCE

This application is a Continuation of U.S. application Ser. No. 17/460,937, filed on Aug. 30, 2021 (now U.S. Pat. No. 11,688,708), the entirety of which is incorporated by reference herein.

BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation. However, these advances have increased the complexity of processing and manufacturing ICs.

In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs.

However, since feature sizes continue to decrease, fabrication processes continue to become more difficult to perform. Therefore, it is a challenge to form reliable semiconductor devices at smaller and smaller sizes.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A-1L are cross-sectional views of various stages of a process for forming a chip structure, in accordance with some embodiments.

FIG. 1G-1 is a top view of the chip structure of FIG. 1G, in accordance with some embodiments.

FIG. 1L-1 is a top view of the chip structure of FIG. 1L, in accordance with some embodiments.

FIG. 2 is a top view of a chip structure, in accordance with some embodiments.

FIG. 3 is a top view of a chip structure, in accordance with some embodiments.

FIG. 4 is a top view of a chip structure, in accordance with some embodiments.

FIG. 5 is a top view of a chip structure, in accordance with some embodiments.

FIG. 6 is a top view of a chip structure, in accordance with some embodiments.

FIG. 7A is a cross-sectional view of a chip structure, in accordance with some embodiments.

FIG. 7B is a top view of the chip structure of FIG. 7A, in accordance with some embodiments.

FIG. 8 is a top view of a chip structure, in accordance with some embodiments.

FIG. 9 is a top view of a chip structure, in accordance with some embodiments.

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FIG. 10 is a top view of a chip structure, in accordance with some embodiments.

FIG. 11 is a top view of a chip structure, in accordance with some embodiments.

FIG. 12 is a top view of a chip structure, in accordance with some embodiments.

FIGS. 13A-13D are cross-sectional views of various stages of a process for forming a chip structure, in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

The term “substantially” in the description, such as in “substantially flat” or in “substantially coplanar”, etc., will be understood by the person skilled in the art. In some embodiments the adjective substantially may be removed. Where applicable, the term “substantially” may also include embodiments with “entirely”, “completely”, “all”, etc. The term “substantially” may be varied in different technologies and be in the deviation range understood by the skilled in the art. For example, the term “substantially” may also relate to 90% of what is specified or higher, such as 95% of what is specified or higher, especially 99% of what is specified or higher, including 100% of what is specified, though the present invention is not limited thereto. Furthermore, terms such as “substantially parallel” or “substantially perpendicular” may be interpreted as not to exclude insignificant deviation from the specified arrangement and may include for example deviations of up to 10°. The word “substantially” does not exclude “completely” e.g. a composition which is “substantially free” from Y may be completely free from Y.

The term “about” may be varied in different technologies and be in the deviation range understood by the skilled in the art. The term “about” in conjunction with a specific distance or size is to be interpreted so as not to exclude insignificant deviation from the specified distance or size. For example, the term “about” may include deviations of up to 10% of what is specified, though the present invention is not limited

thereto. The term “about” in relation to a numerical value x may mean $x \pm 5$ or 10% of what is specified, though the present invention is not limited thereto.

Some embodiments of the disclosure are described. Additional operations can be provided before, during, and/or after the stages described in these embodiments. Some of the stages that are described can be replaced or eliminated for different embodiments. Additional features can be added to the chip structure. Some of the features described below can be replaced or eliminated for different embodiments. Although some embodiments are discussed with operations performed in a particular order, these operations may be performed in another logical order.

FIGS. 1A-1L are cross-sectional views of various stages of a process for forming a chip structure, in accordance with some embodiments. As shown in FIG. 1A, a substrate **110** is provided, in accordance with some embodiments. The substrate **110** includes, for example, a semiconductor wafer (such as a silicon wafer) or a portion of a semiconductor wafer.

In some embodiments, the substrate **110** is made of an elementary semiconductor material including silicon or germanium in a single crystal structure, a polycrystal structure, or an amorphous structure. In some other embodiments, the substrate **110** is made of a compound semiconductor, such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, an alloy semiconductor, such as SiGe or GaAsP, or a combination thereof. The substrate **110** may also include multi-layer semiconductors, semiconductor on insulator (SOI) (such as silicon on insulator or germanium on insulator), or a combination thereof.

In some embodiments, various device elements are formed in and/or over the substrate **110**. The device elements are not shown in figures for the purpose of simplicity and clarity. Examples of the various device elements include active devices, passive devices, other suitable elements, or a combination thereof. The active devices may include transistors or diodes (not shown) formed at a surface of the substrate **110**. The passive devices include resistors, capacitors, or other suitable passive devices.

For example, the transistors may be metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high-voltage transistors, high-frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), etc. Various processes, such as front-end-of-line (FEOL) semiconductor fabrication processes, are performed to form the various device elements. The FEOL semiconductor fabrication processes may include deposition, etching, implantation, photolithography, annealing, planarization, one or more other applicable processes, or a combination thereof.

In some embodiments, isolation features (not shown) are formed in the substrate **110**. The isolation features are used to surround active regions and electrically isolate various device elements formed in and/or over the substrate **110** in the active regions. In some embodiments, the isolation features include shallow trench isolation (STI) features, local oxidation of silicon (LOCOS) features, other suitable isolation features, or a combination thereof.

As shown in FIG. 1A, an interconnect structure **120** is formed over the substrate **110**, in accordance with some embodiments. The interconnect structure **120** includes a dielectric structure **122**, wiring layers **124**, and conductive vias **126**, in accordance with some embodiments. The dielectric structure **122** is formed over a surface **112** of the substrate **110**, in accordance with some embodiments.

The wiring layers **124** and the conductive vias **126** are formed in the dielectric structure **122**, in accordance with some embodiments. The conductive vias **126** are electrically connected between different wiring layers **124** and between the wiring layer **124** and the aforementioned device elements, in accordance with some embodiments.

The wiring layers **124** include top metal wiring layers **124a** and **124b** and wiring layers **124c**, in accordance with some embodiments. Both of the top metal wiring layers **124a** and **124b** are thicker than the wiring layers **124c**, in accordance with some embodiments.

The top metal wiring layer **124a** has a thickness T_{124a} ranging from about 0.6 μm to about 1 μm , in accordance with some embodiments. The top metal wiring layer **124b** has a thickness T_{124b} ranging from about 0.6 μm to about 1 μm , in accordance with some embodiments. The wiring layer **124c** has a thickness T_{124c} ranging from about 0.04 μm to about 0.5 μm , in accordance with some embodiments.

Since both of the top metal wiring layers **124a** and **124b** are thicker than the wiring layers **124c**, the top metal wiring layers **124a** and **124b** are able to withstand greater bonding stress in a subsequent bonding process than the wiring layers **124c** and able to suppress stress migration to the wiring layers **124c** therebelow, in accordance with some embodiments.

In some embodiments, a ratio of an area of a top surface of the top metal wiring layer **124a** or **124b** to an area of a top surface **121** of the interconnect structure **120** is substantially equal to or greater than 20%. The ratio is also referred to as a density of the top metal wiring layer **124a** or **124b**, in accordance with some embodiments. If the ratio (i.e. the density of the top metal wiring layer **124a** or **124b**) is less than 20%, the top metal wiring layers **124a** and **124b** may be unable to suppress stress migration to the wiring layers **124c** therebelow, in accordance with some embodiments.

The dielectric structure **122** is made of an oxide-containing material (e.g. silicon oxide or undoped silicate glass) or another suitable insulating material, in accordance with some embodiments. The wiring layers **124** and the conductive vias **126** are made of conductive materials such as metal (e.g., aluminum, copper or tungsten) or alloys thereof, in accordance with some embodiments.

As shown in FIG. 1A, a passivation layer **130** is formed over the interconnect structure **120**, in accordance with some embodiments. The passivation layer **130** is used as an anti-acid layer to prevent acid (used in subsequent processes) from penetrating into the interconnect structure **120**, in accordance with some embodiments.

The passivation layer **130** is made of a dielectric material, such as an oxide-containing material (e.g., silicon oxide or undoped silicate glass (USG)), in accordance with some embodiments. The passivation layer **130** is formed using a deposition process (e.g., a chemical vapor deposition process or a physical vapor deposition process), in accordance with some embodiments.

As shown in FIG. 1A, a metal-insulator-metal (MIM) capacitor **140** is formed over the passivation layer **130**, in accordance with some embodiments. The MIM capacitor **140** includes a bottom metal layer (not shown), an insulating layer (not shown), and a top metal layer (not shown), in accordance with some embodiments. The insulating layer is sandwiched between the bottom metal layer and the top metal layer, in accordance with some embodiments.

The bottom metal layer and the top metal layer are made of titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta₂N₃), copper (Cu), copper alloy, aluminum (Al), aluminum (Al) alloy, copper aluminum alloy (AlCu),

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tungsten (W), or tungsten (W) alloy, in accordance with some embodiments. The bottom metal layer and the top metal layer are formed by a procedure including depositing, photolithography, and etching processes.

The deposition processes include chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or applicable methods. The photolithography processes include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, and drying (e.g., hard baking), in accordance with some embodiments. The etching processes include dry etching, wet etching, and/or other etching methods.

The insulating layer is made of dielectric materials, such as silicon oxide, silicon nitride or silicon glass. In some embodiments, the insulating layer is formed by a chemical vapor deposition (CVD) process or physical vapor deposition (PVD) process.

As shown in FIG. 1A, a passivation layer 150 is formed over the passivation layer 130 and the MIM capacitor 140, in accordance with some embodiments. The passivation layer 150 is used as a waterproof layer to prevent water from penetrating into the interconnect structure 120, in accordance with some embodiments.

The passivation layer 150 is made of a dielectric material, such as a nitride-containing material (e.g., silicon nitride or silicon oxynitride), in accordance with some embodiments. The passivation layer 150 is formed using a deposition process (e.g., a chemical vapor deposition process or a physical vapor deposition process), in accordance with some embodiments.

As shown in FIG. 1B, portions of the passivation layers 130 and 150 are removed to form a through hole TH1 in the passivation layers 130 and 150, in accordance with some embodiments. In some embodiments, the through hole TH1 further extends into the top metal wiring layer 124b. The through hole TH1 exposes a portion of the top metal wiring layer 124b, in accordance with some embodiments. The removal process includes a photolithography process and an etching process, such as a dry etching process, in accordance with some embodiments.

Thereafter, a barrier layer (not shown) is conformally formed over the passivation layers 130 and 150 and in the through hole TH1, in accordance with some embodiments. The barrier layer is made of nitrides such as tantalum nitride (TaN), in accordance with some embodiments. The barrier layer is formed using a deposition process, such as a physical vapor deposition process, in accordance with some embodiments.

As shown in FIG. 1C, a seed layer 160 is conformally formed over the barrier layer (not shown), in accordance with some embodiments. In some embodiments, the barrier layer is not formed. The seed layer 160 conformally covers a bottom surface B1 and inner walls N1 of the through hole TH1, in accordance with some embodiments.

The seed layer 160 is made of a conductive material, such as metal (e.g., copper, aluminum, gold, silver, or tungsten) or alloys thereof, in accordance with some embodiments. The seed layer 160 is formed using a deposition process, such as a physical vapor deposition process, in accordance with some embodiments.

As shown in FIG. 1D, a mask layer 170 is formed over the seed layer 160, in accordance with some embodiments. The mask layer 170 has trenches 172 and 174 exposing portions of the seed layer 160, in accordance with some embodiments. The trench 172 exposes a portion of the seed layer 160 in the through hole TH1 and a portion of the seed layer

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160 over a top surface 152 of the passivation layer 150, in accordance with some embodiments. The trench 172 is wider than the trench 174, in accordance with some embodiments.

The trenches 172 and 174 have a trapezoid-like shape, which has a narrow top and a wide bottom, and therefore conductive lines subsequently formed therein have a trapezoid-like shape as well, which prevents the conductive lines from collapsing and therefore improves the stability of the conductive lines, in accordance with some embodiments. The mask layer 170 is made of a polymer material, such as a photoresist material, in accordance with some embodiments.

After the mask layer 170 is formed, a descum process is performed over the seed layer 160 exposed by the trenches 172 and 174 to remove the residues thereover, in accordance with some embodiments. The descum process includes an etching process such as a plasma etching process, in accordance with some embodiments.

As shown in FIG. 1E, a conductive layer 180 is formed over the seed layer 160 exposed by the trenches 172 and 174, in accordance with some embodiments. The conductive layer 180 is made of a conductive material, such as metal (e.g., copper) or alloys thereof, in accordance with some embodiments. The conductive layer 180 is formed by a plating process, such as an electroplating process, in accordance with some embodiments.

As shown in FIG. 1F, the mask layer 170 is removed, in accordance with some embodiments. As shown in FIGS. 1E and 1G, the seed layer 160 originally under the mask layer 170 is removed, in accordance with some embodiments. The removal process includes an etching process such as a wet etching process, in accordance with some embodiments. Thereafter, the barrier layer (not shown), which is not covered by the conductive layer 180, is removed, in accordance with some embodiments. The removal process includes an etching process such as a dry etching process, in accordance with some embodiments.

The removal process for removing the seed layer 160 originally under the mask layer 170 may further remove portions of the passivation layer 150, which are not covered by the conductive layer 180. Therefore, recesses 154 are formed in the passivation layer 150, in accordance with some embodiments. The passivation layers 130 and 150 together form a first passivation layer PA1, in accordance with some embodiments. The first passivation layer PA1 has a thickness T_{PA1} ranging from about 0.2 μm to about 0.8 μm , in accordance with some embodiments.

FIG. 1G-1 is a top view of the chip structure of FIG. 1G, in accordance with some embodiments. As shown in FIGS. 1G and 1G-1, the conductive layer 180 in the through hole TH1 and the seed layer 160 thereunder together form a conductive via structure 182, in accordance with some embodiments. The conductive via structure 182 pass through the passivation layers 130 and 150, in accordance with some embodiments. In some embodiments, the conductive via structure 182 have an inverted trapezoid shape.

As shown in FIGS. 1E, 1G, and 1G-1, the conductive layer 180, originally in the trench 172, and the seed layer 160 thereunder together form a conductive line 184, in accordance with some embodiments. The conductive line 184 has a linewidth W_{184} ranging from about 10 μm to about 50 μm , in accordance with some embodiments. The conductive via structure 182 is directly connected between the conductive line 184 and the top metal wiring layer 124b thereunder, in accordance with some embodiments.

The conductive layer **180**, originally in the trenches **174**, and the seed layer **160** thereunder together form conductive lines **186**, in accordance with some embodiments. The conductive line **186** has a linewidth W_{186} ranging from about 2 μm to about 50 μm , in accordance with some embodiments.

In some embodiments, the linewidth W_{184} is greater than the linewidth W_{186} , which increase the alignment tolerance between the conductive line **184** and a conductive pillar subsequently formed thereon. In some embodiments, a ratio of the linewidth W_{184} to the linewidth W_{186} ranges from about 1.5 to about 2.5. If the ratio is less than 1.5, the alignment tolerance between the conductive line **184** and the conductive pillar may be unable to be increased. If the ratio is greater than 2.5, the conductive line **184** may occupy too much layout space. In some other embodiments, the linewidth W_{184} is substantially equal to the linewidth W_{186} , which reserves more space for wiring layout.

The conductive lines **184** and **186** together form a wiring layer **180R**, in accordance with some embodiments. The wiring layer **180R** is thicker than the wiring layers **124**, in accordance with some embodiments. The wiring layer **180R** has a thickness T_{180R} ranging from about 2 μm to about 10 μm , in accordance with some embodiments. The conductive lines **184** and **186** are spaced apart from each other by gaps **G1**, in accordance with some embodiments.

The conductive line **184** has a top surface **184s1** and a lower surface **184s2**, in accordance with some embodiments. Each conductive line **186** has a top surface **186s1** and a lower surface **186s2**, in accordance with some embodiments. The top surface **184s1** is substantially level with (or coplanar with) the top surfaces **186s1** of the conductive lines **186**, in accordance with some embodiments. The lower surface **184s2** is substantially level with (or coplanar with) the lower surfaces **186s2** of the conductive lines **186**, in accordance with some embodiments.

As shown in FIG. 1G-1, the conductive lines **184** and **186** are substantially parallel to each other, in accordance with some embodiments. In some embodiments, a distance **D1** between the conductive lines **184** and **186** is substantially equal to a distance **D2** between the conductive lines **186**, which improves the arrangement uniformity of the wiring layer **180R** and therefore improves the planarity of a top surface of an insulating layer that is subsequently formed thereover. Therefore, the planarity of a top surface of a conductive pillar subsequently formed over the insulating layer is improved as well, in accordance with some embodiments. The distance **D1** ranges from about 4 μm to about 30 μm , in accordance with some embodiments. The distance **D2** ranges from about 4 μm to about 30 μm , in accordance with some embodiments.

In some embodiments, a ratio of an area of a top surface of the wiring layer **180R** to the area of the top surface **121** of the interconnect structure **120** ranges from about 55% to about 70%. The ratio is also referred to as a density of the wiring layer **180R**, in accordance with some embodiments.

If the ratio (i.e. the density of the wiring layer **180R**) is less than 55%, the arrangement uniformity of the wiring layer **180R** is not enough, which is not conducive to the planarity of a top surface of an insulating layer that is subsequently formed thereover. If the ratio (i.e. the density of the wiring layer **180R**) is greater than 70%, the thermal stress between the wiring layer **180R** and the insulating layer subsequently formed thereover may be large.

As shown in FIG. 1H, a passivation layer **190** is conformally formed over the passivation layer **150** and the wiring layer **180R**, in accordance with some embodiments. The

passivation layer **190** conformally covers the conductive lines **184** and **186** and the gaps **G1** therebetween, in accordance with some embodiments. The passivation layer **190** is thicker than the first passivation layer **PA1**, in accordance with some embodiments. The passivation layer **190** has a thickness T_{190} ranging from about 0.8 μm to about 1.7 μm , in accordance with some embodiments.

The passivation layer **190** is made of a dielectric material, such as nitrides (e.g., silicon nitride or silicon oxynitride), in accordance with some embodiments. The passivation layer **190** is formed using a deposition process (e.g., a chemical vapor deposition process or a physical vapor deposition process), in accordance with some embodiments.

As shown in FIG. 1H, an insulating layer **210** is formed over the passivation layer **190**, in accordance with some embodiments. The gaps **G1** are filled with the insulating layer **210**, in accordance with some embodiments. The insulating layer **210** has a substantially flat top surface **212**, in accordance with some embodiments. Since the arrangement uniformity of the wiring layer **180R** is improved, the planarity of the top surface **212** is improved as well, in accordance with some embodiments.

The top surface **212** continuously extends across the conductive lines **184** and **186** and the gaps **G1** therebetween, in accordance with some embodiments. The insulating layer **210** has a thickness T_{210} ranging from about 5 μm to about 15 μm , in accordance with some embodiments. The insulating layer **210** is made of a polymer material such as polyimide (PI), in accordance with some embodiments.

As shown in FIG. 1I, portions of the insulating layer **210** and the passivation layer **190** over the conductive line **184** are removed to form a through hole **TH2** in the insulating layer **210** and the passivation layer **190**, in accordance with some embodiments. The through hole **TH2** exposes a portion of the conductive line **184**, in accordance with some embodiments. The removal process includes an etching process such as a dry etching process, in accordance with some embodiments.

Thereafter, a seed layer (not shown) is conformally formed over the passivation layer **190**, the insulating layer **210**, and the conductive line **184**, in accordance with some embodiments. The seed layer is in direct contact with the passivation layer **190**, the insulating layer **210**, and the conductive line **184**, in accordance with some embodiments.

The seed layer is made of a conductive material, such as metal (e.g., copper, aluminum, gold, silver, or tungsten) or alloys thereof, in accordance with some embodiments. The seed layer is formed using a deposition process, such as a physical vapor deposition process, in accordance with some embodiments.

As shown in FIG. 1J, a mask layer **230** is formed over the seed layer (not shown), in accordance with some embodiments. The mask layer **230** has an opening **232** exposing a portion of the seed layer, in accordance with some embodiments. The mask layer **230** is made of a polymer material, such as a photoresist material, in accordance with some embodiments.

As shown in FIG. 1J, a conductive pillar **240** is formed over the seed layer exposed by the opening **232**, in accordance with some embodiments. Since the seed layer is very thin, the seed layer may be seen as a portion of the conductive pillar **240**, in accordance with some embodiments. The conductive pillar **240** has a top surface **243**, in accordance with some embodiments. Since the conductive pillar **240** is formed over the substantially flat top surface

212 of the insulating layer **210**, the top surface **243** is a substantially flat top surface, in accordance with some embodiments.

The conductive pillar **240** is made of a conductive material, such as metal (e.g., titanium, copper, nickel, or aluminum) or alloys thereof, in accordance with some embodiments. The conductive pillar **240** is formed using a plating process, such as an electroplating process, in accordance with some embodiments.

As shown in FIG. 1J, a solder layer **250a** is formed over the conductive pillar **240**, in accordance with some embodiments. The solder layer **250a** is made of a conductive material, such as metal (e.g., tin or the like) or alloys thereof, in accordance with some embodiments. The solder layer **250a** is formed using a plating process, such as an electroplating process, in accordance with some embodiments.

As shown in FIG. 1K, the mask layer **230** is removed, in accordance with some embodiments. The seed layer originally under the mask layer **230** is removed as well, in accordance with some embodiments. The removal process includes an etching process such as a wet etching process, in accordance with some embodiments.

The conductive pillar **240** is formed over the insulating layer **210** and in the through hole TH2, in accordance with some embodiments. The conductive pillar **240** is over the conductive lines **184** and **186**, in accordance with some embodiments. The conductive pillar **240** is formed in one piece, in accordance with some embodiments.

The conductive pillar **240** has a bottom protruding portion **242** and a lower surface **244**, in accordance with some embodiments. The bottom protruding portion **242** protrudes from the lower surface **244**, in accordance with some embodiments. The bottom protruding portion **242** is in the through hole TH2, in accordance with some embodiments.

The bottom protruding portion **242** passes through the insulating layer **210** and the passivation layer **190**, in accordance with some embodiments. The bottom protruding portion **242** is in direct contact with the conductive line **184**, in accordance with some embodiments. The conductive via structure **182** is under the bottom protruding portion **242**, which shortens the conductive path between the conductive pillar **240** and the wiring layer **124b**, in accordance with some embodiments.

In some embodiments, the linewidth W_{184} of the conductive line **184** under the conductive pillar **240** is less than a width W_{240} of the conductive pillar **240**. The width W_{240} ranges from about 60 μm to about 105 μm , in accordance with some embodiments. The conductive pillar **240** has a thickness T_{240} ranges from about 30 μm to about 50 μm , in accordance with some embodiments.

FIG. 1L-1 is a top view of the chip structure of FIG. 1L, in accordance with some embodiments. As shown in FIGS. 1K, 1L and 1L-1, a reflow process is performed over the solder layer **250a** to form a solder bump **250**, in accordance with some embodiments. The solder bump **250** is in direct contact with the conductive pillar **240**, in accordance with some embodiments. In this step, a chip structure **100** is substantially formed, in accordance with some embodiments.

As shown in FIG. 1L-1, the through hole TH2 and the bottom protruding portion **242** have a strip shape, such as a rectangle shape, in accordance with some embodiments. In some embodiments, a longitudinal axis A1 of the through hole TH2 or the bottom protruding portion **242** is substantially parallel to a portion of the conductive line **184** under the conductive pillar **240**.

As shown in FIG. 1L-1, a width W_{242} of the bottom protruding portion **242** or the through hole TH2 ranges from about 5 μm to about 15 μm , in accordance with some embodiments. As shown in FIG. 1L-1, a length L_{242} of the bottom protruding portion **242** or the through hole TH2 ranges from about 20 μm to about 40 μm , in accordance with some embodiments.

As shown in FIG. 1L-1, a width W_{182} of the conductive via structure **182** or the through hole TH1 ranges from about 1 μm to about 6 μm , in accordance with some embodiments.

As shown in FIG. 1L-1, a length L_{182} of the conductive via structure **182** or the through hole TH1 ranges from about 1 μm to about 6 μm , in accordance with some embodiments.

As shown in FIG. 1L-1, a distance D3 between the conductive via structures **182** is substantially equal to or greater than 2 μm , in accordance with some embodiments.

As shown in FIG. 1L, the conductive via structure **182**, the bottom protruding portion **242**, and a center portion **241** of the conductive pillar **240** are aligned with each other in a direction B perpendicular to the top surface **112** of the substrate **110**, in accordance with some embodiments.

The material property of copper may reduce the stress migration and the electromigration effect, in accordance with some embodiments. Therefore, if the conductive line **184** and the conductive via structure **182** are made of copper, the stress migration and the electromigration effect are reduced, in accordance with some embodiments.

The (thick) insulating layer **210** is able to absorb a portion of the bonding stress in a subsequent bonding process, which reduces the bonding stress transmitted to the wiring layers **180R** and **124** therebelow, in accordance with some embodiments. Therefore, there is no need to form conductive pads in the wiring layer **180R** under the conductive pillar **240**, in accordance with some embodiments. Therefore, the conductive pad is replaced by the conductive line **184**, which is narrower than the conductive pad, in accordance with some embodiments. As a result, the (narrower) conductive line **184** may reserve more space under the conductive pillar **240** for wiring layout than the conductive pads, in accordance with some embodiments.

FIG. 2 is a top view of a chip structure **200**, in accordance with some embodiments. As shown in FIG. 2, the chip structure **200** is similar to the chip structure **100** of FIGS. 1L and 1L-1, except that the conductive via structure **182**, the conductive line **184**, and the bottom protruding portion **242** of the conductive pillar **240** are not under the center portion **241** of the conductive pillar **240**, in accordance with some embodiments. That is, the conductive via structure **182**, the conductive line **184**, and the bottom protruding portion **242** of the conductive pillar **240** are misaligned with the center portion **241** of the conductive pillar **240**, in accordance with some embodiments.

FIG. 3 is a top view of a chip structure **300**, in accordance with some embodiments. As shown in FIG. 3, the chip structure **300** is similar to the chip structure **100** of FIGS. 1L and 1L-1, except that the conductive line **184**, which is under the conductive pillar **240**, is wider than the conductive line **184**, which is not under the conductive pillar **240**, in accordance with some embodiments.

That is, the conductive line **184** has a wide portion **184w** and narrow portions **184n1** and **184n2**, in accordance with some embodiments. The wide portion **184w** is wider than both of the narrow portions **184n1** and **184n2**, in accordance with some embodiments. The wide portion **184w** is connected between the narrow portions **184n1** and **184n2**, in accordance with some embodiments. The wide portion **184w** is under the conductive pillar **240**, in accordance with some

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embodiments. The narrow portions **184n1** and **184n2** are not under the conductive pillar **240**, in accordance with some embodiments.

In some embodiments, a linewidth W_{184w} of the wide portion **184w** is greater than a linewidth W_{184n1} of the narrow portion **184n1**. In some embodiments, the linewidth W_{184w} of the wide portion **184w** is greater than a linewidth W_{184n2} of the narrow portion **184n2**. Therefore, the wide portion **184w** may increase the alignment tolerance between the conductive line **184** and the bottom protruding portion **242** of the conductive pillar **240**, and the narrow portions **184n1** and **184n2** may reserve more space, which is not under the conductive pillar **240**, for wiring layout.

In some embodiments, a ratio of the linewidth W_{184w} to the linewidth W_{184n1} ranges from about 1.5 to about 2.5. In some embodiments, a ratio of the linewidth W_{184w} to the linewidth W_{184n2} ranges from about 1.5 to about 2.5.

In some embodiments, the linewidth W_{184w} is greater than the linewidth W_{186} of the conductive lines **186**. In some embodiments, the linewidth W_{184n1} or W_{184n2} is substantially equal to the linewidth W_{186} of the conductive lines **186**.

FIG. 4 is a top view of a chip structure **400**, in accordance with some embodiments. As shown in FIG. 4, the chip structure **400** is similar to the chip structure **100** of FIGS. 1L and 1L-1, except that portions **184b** and **184c** of the conductive line **184**, which are not under the conductive pillar **240**, are not parallel to portions **186b** and **186c** of the conductive line **186**, which are not under the conductive pillar **240**, in accordance with some embodiments.

Specifically, the conductive line **184** includes portions **184a**, **184b** and **184c**, in accordance with some embodiments. The portion **184a** is connected between the portions **184b** and **184c** and is under the conductive pillar **240**, in accordance with some embodiments. The portions **184b** and **184c** are not under the conductive pillar **240**, in accordance with some embodiments. The portions **184b** and **184c** are not parallel to the portion **184a**, in accordance with some embodiments.

The conductive line **186** includes portions **186a**, **186b** and **186c**, in accordance with some embodiments. The portion **186a** is connected between the portions **186b** and **186c** and is under the conductive pillar **240**, in accordance with some embodiments. The portions **186b** and **186c** are not under the conductive pillar **240**, in accordance with some embodiments. The portions **186b** and **186c** are not parallel to the portion **186a**, in accordance with some embodiments.

The portion **184a** of the conductive line **184** is parallel to the portion **186a** of the conductive line **186**, in accordance with some embodiments. The portions **184b** and **184c** are not parallel to the portions **186b** and **186c**, in accordance with some embodiments. That is, the conductive lines **184** and **186** under the conductive pillar **240** are parallel to each other to form the insulating layer **210** with the substantially flat top surface so as to form the conductive pillar **240** with the substantially flat top surface. The design of the conductive lines **184** and **186**, which are not under the conductive pillar **240**, does not affect the planarity of the top surface of the conductive pillar **240** and therefore has more freedom of wiring layout.

FIG. 5 is a top view of a chip structure **500**, in accordance with some embodiments. As shown in FIG. 5, the chip structure **500** is similar to the chip structure **100** of FIGS. 1L and 1L-1, except that the bottom protruding portion **242** of the conductive pillar **240** of the chip structure **500** has a substantially oval-like shape, in accordance with some embodiments. In some embodiments, opposite sidewalls

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242s1 and **242s2** of the bottom protruding portion **242** are substantially parallel to each other.

FIG. 6 is a top view of a chip structure **600**, in accordance with some embodiments. As shown in FIG. 6, the chip structure **600** is similar to the chip structure **100** of FIGS. 1L and 1L-1, except that the bottom protruding portion **242** of the conductive pillar **240** of the chip structure **600** has an oval shape, in accordance with some embodiments.

FIG. 7A is a cross-sectional view of a chip structure **700**, in accordance with some embodiments. FIG. 7B is a top view of the chip structure **700** of FIG. 7A, in accordance with some embodiments. As shown in FIGS. 7A and 7B, the chip structure **700** is similar to the chip structure **100** of FIGS. 1L and 1L-1, except that a width W_{242t} of a top **242t** of the bottom protruding portion **242** of the conductive pillar **240** (or a top **TH2t** of the through hole **TH2**) is greater than the linewidth W_{184} of the conductive line **184**, in accordance with some embodiments.

The width W_{242} of a bottom **242b** of the bottom protruding portion **242** of the conductive pillar **240** (or a bottom **TH2b** of the through hole **TH2**) is less than the linewidth W_{184} of the conductive line **184**, in accordance with some embodiments.

FIG. 8 is a top view of a chip structure **800**, in accordance with some embodiments. As shown in FIG. 8, the chip structure **800** is similar to the chip structure **100** of FIGS. 1L and 1L-1, except that the conductive pillar **240** has more than one bottom protruding portion **242**, in accordance with some embodiments. The conductive pillar **240** has, for example, three bottom protruding portions **242**, in accordance with some embodiments. The bottom protruding portions **242** are arranged along a portion of the conductive line **184** under the conductive pillar **240**, in accordance with some embodiments.

The bottom protruding portions **242** are able to share the bonding stress during a subsequent bonding process for bonding the conductive pillar **240** to a substrate (not shown) through the solder bump **250**, in accordance with some embodiments. Therefore, the bottom protruding portions **242** are able to prevent the bonding stress from concentrating in only one bottom protruding portion **242**, which improves the reliability of the chip structure **800**, in accordance with some embodiments.

If one of the bottom protruding portions **242** is broken, the others of the bottom protruding portions **242** may still be connected to the conductive line **184**. Therefore, the multiple bottom protruding portions **242** may improve the reliability of the electrical connection between the conductive line **184** and the conductive pillar **240**.

FIG. 9 is a top view of a chip structure **900**, in accordance with some embodiments. As shown in FIG. 9, the chip structure **900** is similar to the chip structure **800** of FIG. 8, except that the bottom protruding portions **242** of the conductive pillar **240** of the chip structure **900** have different shapes, in accordance with some embodiments. In some embodiments, the bottom protruding portions **242** have a square-like shape, a substantially oval-like shape, and a round-like shape, respectively, in accordance with some embodiments.

FIG. 10 is a top view of a chip structure **1000**, in accordance with some embodiments. As shown in FIG. 10, the chip structure **1000** is similar to the chip structure **100** of FIGS. 1L and 1L-1, except that the conductive lines **184** and **186** under the conductive pillar **240** and the bottom protruding portion **242** of the conductive pillar **240** have an L-like shape, in accordance with some embodiments.

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FIG. 11 is a top view of a chip structure 1100, in accordance with some embodiments. As shown in FIG. 11, the chip structure 1100 is similar to the chip structure 1000 of FIG. 10, except that the conductive pillar 240 has two bottom protruding portions 242, which have a straight strip shape such as a rectangle shape, in accordance with some embodiments.

FIG. 12 is a top view of a chip structure 1200, in accordance with some embodiments. As shown in FIG. 12, the chip structure 1200 is similar to the chip structure 100 of FIGS. 1L and 1L-1, except that the chip structure 1200 has more than one conductive line 184 and more than one bottom protruding portion 242, in accordance with some embodiments. Specifically, the conductive pillar 240 has two bottom protruding portions 242, which are respectively connected to two conductive lines 184 thereunder, in accordance with some embodiments.

FIGS. 13A-13D are cross-sectional views of various stages of a process for forming a chip structure 1300, in accordance with some embodiments. As shown in FIG. 13A, after the step of the formation of the passivation layer 190 of FIG. 1H, a portion of the passivation layer 190 is removed to form an opening 192 in the passivation layer 190, in accordance with some embodiments. The opening 192 exposes a portion of the conductive line 184, in accordance with some embodiments. The removal process includes a photolithography process and an etching process (e.g., a dry etching process), in accordance with some embodiments.

As shown in FIG. 13B, an insulating layer 210 is formed over the passivation layer 190, in accordance with some embodiments. The opening 192 is filled with the insulating layer 210, in accordance with some embodiments. The insulating layer 210 has a substantially flat top surface 212, in accordance with some embodiments.

As shown in FIG. 13C, a portion of the insulating layer 210 is removed to form an opening 214 in the insulating layer 210 and communicating with the opening 192 thereunder, in accordance with some embodiments. The removal process further removes a portion of the passivation layer 190 under the opening 214, in accordance with some embodiments. The opening 214 is wider than the opening 192, in accordance with some embodiments. The removal process includes a photolithography process and an etching process (e.g., a dry etching process), in accordance with some embodiments.

As shown in FIG. 13D, the steps of FIGS. 1J-1L are performed to form the conductive pillar 240 and the solder bump 250 over the conductive line 184 and the insulating layer 210, in accordance with some embodiments. In this step, a chip structure 1300 is substantially formed, in accordance with some embodiments.

Processes and materials for forming the semiconductor structures 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200 and 1300 may be similar to, or the same as, those for forming the semiconductor structure 100 described above.

In accordance with some embodiments, chip structures and methods for forming the same are provided. The methods (for forming the chip structure) form a conductive pillar directly on a conductive line, not a conductive pad, to reserve more space under the conductive pillar for wiring layout.

In accordance with some embodiments, a chip structure is provided. The chip structure includes a substrate. The chip structure includes a first conductive line over the substrate. The chip structure includes an insulating layer over the substrate and the first conductive line. The chip structure

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includes a conductive pillar over and passing through the insulating layer. The conductive pillar is formed in one piece, the conductive pillar is in direct contact with the first conductive line, and a first sidewall of the first conductive line extends across a second sidewall of the conductive pillar in a top view of the first conductive line and the conductive pillar. The chip structure includes a solder bump on the conductive pillar. The solder bump is in direct contact with the conductive pillar.

In accordance with some embodiments, a chip structure is provided. The chip structure includes a substrate. The chip structure includes a first conductive line and a second conductive line over the substrate and laterally spaced apart from each other. The chip structure includes a conductive pillar over the first conductive line and the second conductive line. The conductive pillar is formed in one piece, and the conductive pillar is in direct contact with the first conductive line. The chip structure includes a solder bump on the conductive pillar. The solder bump is in direct contact with the conductive pillar.

In accordance with some embodiments, a method for forming a chip structure is provided. The method includes forming a first conductive line over a substrate. The method includes forming an insulating layer over the substrate and the first conductive line. The insulating layer has a through hole exposing a first portion of the first conductive line. The method includes forming a conductive pillar over the insulating layer and in the through hole. The conductive pillar is formed in one piece, the conductive pillar is in direct contact with the first conductive line, and a first sidewall of the first conductive line extends across a second sidewall of the conductive pillar in a top view of the first conductive line and the conductive pillar. The method includes forming a solder bump on the conductive pillar. The solder bump is in direct contact with the conductive pillar.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A chip structure, comprising:

a substrate;

a first conductive line over the substrate;

an insulating layer over the substrate and the first conductive line;

a conductive pillar over and passing through the insulating layer, wherein the conductive pillar is formed in one piece, the conductive pillar extends to the first conductive line, a first sidewall of the first conductive line extends across a second sidewall of the conductive pillar in a top view of the first conductive line and the conductive pillar, the conductive pillar vertically overlaps a first portion of the first conductive line, a first linewidth of the first portion of the first conductive line is less than a width of the conductive pillar, and the first linewidth and the width are measured along an axis perpendicular to a first longitudinal axis of the first portion of the first conductive line; and

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a solder bump on the conductive pillar, wherein a bottom surface of the solder bump is substantially level with a top surface of the conductive pillar.

2. The chip structure as claimed in claim 1, wherein the conductive pillar has a lower surface and a bottom protruding portion protruding from the lower surface, the bottom protruding portion passes through the insulating layer over the first conductive line, and the bottom protruding portion has a rectangular shape in the top view of the conductive pillar and the first conductive line.

3. The chip structure as claimed in claim 2, wherein a second longitudinal axis of the bottom protruding portion is substantially parallel to the first sidewall of the first conductive line under the conductive pillar in the top view of the conductive pillar and the first conductive line.

4. The chip structure as claimed in claim 1, further comprising:

a second conductive line between the substrate and the conductive pillar, wherein the first conductive line is laterally spaced apart from the second conductive line, and the conductive pillar vertically overlaps the second conductive line.

5. The chip structure as claimed in claim 4, wherein the first linewidth of the first portion of the first conductive line is greater than a second linewidth of the second conductive line, and the second linewidth is measured along the axis perpendicular to the first longitudinal axis of the first portion of the first conductive line.

6. The chip structure as claimed in claim 5, wherein the first linewidth of the first portion of the first conductive line covered by the conductive pillar is greater than a third linewidth of a second portion of the first conductive line not covered by the conductive pillar, and the third linewidth is measured along the axis perpendicular to the first longitudinal axis of the first portion of the first conductive line.

7. The chip structure as claimed in claim 4, wherein the first conductive line under the conductive pillar is substantially parallel to the second conductive line under the conductive pillar.

8. The chip structure as claimed in claim 1, wherein the first conductive line is spaced apart from the second conductive line by a gap, the gap is filled with the insulating layer, and the insulating layer has a substantially flat top surface continuously extending across the first conductive line, the gap, and the second conductive line.

9. A chip structure, comprising:

a substrate;

a first conductive line over the substrate;

an insulating layer over the substrate and the first conductive line;

a conductive pillar over and passing through the insulating layer, wherein the conductive pillar is formed in one piece, the conductive pillar extends to the first conductive line, a first sidewall of the first conductive line extends across a second sidewall of the conductive pillar in a top view of the first conductive line and the conductive pillar, the conductive pillar vertically overlaps a first portion of the first conductive line, a first linewidth of the first portion of the first conductive line is less than a width of the conductive pillar, and the first linewidth and the width are measured along an axis perpendicular to a longitudinal axis of the first portion of the first conductive line;

a solder bump on the conductive pillar; and

a second conductive line between the substrate and the conductive pillar, wherein the conductive pillar vertically overlaps the second conductive line, the first

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conductive line is laterally spaced apart from the second conductive line, and a first top surface of the first conductive line is substantially level with a second top surface of the second conductive line.

10. The chip structure as claimed in claim 9, wherein the first linewidth of the first portion of the first conductive line is greater than a second linewidth of the second conductive line, and the second linewidth is measured along the axis perpendicular to the longitudinal axis of the first portion of the first conductive line.

11. The chip structure as claimed in claim 10, wherein a distance between the first conductive line and the second conductive line decreases toward the substrate.

12. The chip structure as claimed in claim 9, wherein the conductive pillar has a lower surface and a bottom protruding portion protruding from the lower surface, the bottom protruding portion passes through the insulating layer over the first conductive line, and the bottom protruding portion has four square corners in the top view of the conductive pillar and the first conductive line.

13. The chip structure as claimed in claim 9, wherein the first conductive line further has a second portion not covered by the conductive pillar, a first distance between the first portion of the first conductive line and the second conductive line is less than a second distance between the second portion and the second conductive line.

14. The chip structure as claimed in claim 13, wherein the first conductive line has a first recess and a second recess, and the second portion is between the first recess and the second recess.

15. A chip structure, comprising:

a substrate;

a first conductive line over the substrate;

an insulating layer over the substrate and the first conductive line;

a conductive pillar over and passing through the insulating layer, wherein the conductive pillar is formed in one piece, the conductive pillar extends to the first conductive line, a first sidewall of the first conductive line extends across a second sidewall of the conductive pillar in a top view of the first conductive line and the conductive pillar, the conductive pillar vertically overlaps a first portion of the first conductive line, a first linewidth of the first portion of the first conductive line is less than a width of the conductive pillar, the first linewidth and the width are measured along an axis perpendicular to a longitudinal axis of the first portion of the first conductive line, and a second portion of the insulating layer between the conductive pillar and the substrate has a comb shape in a cross-sectional view of the insulating layer; and

a solder bump on the conductive pillar, wherein a bottom surface of the solder bump is substantially level with a top surface of the conductive pillar.

16. The chip structure as claimed in claim 15, wherein the first conductive line is embedded in the second portion of the insulating layer.

17. The chip structure as claimed in claim 16, further comprising:

a second conductive line between the substrate and the conductive pillar, wherein the first conductive line is laterally spaced apart from the second conductive line, and the conductive pillar vertically overlaps the second conductive line.

18. The chip structure as claimed in claim 17, wherein the second conductive line is embedded in the second portion of the insulating layer.

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19. The chip structure as claimed in claim **15**, wherein the insulating layer between the first conductive line and the second conductive line has an inverted trapezoid shape.

20. The chip structure as claimed in claim **15**, wherein the first conductive line further has a first recess and a second recess, the conductive pillar does not cover the first recess and the second recess, and the conductive pillar is between the first recess and the second recess.

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